DEGREE PROJECT

AI based Student Wellbeing & Performance Monitoring Toolkit for EdTech

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Master Programme in Applied AI
2026

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Statement of Originality

We hereby declare that this thesis is our own work and has not been submitted in any form for another degree or diploma at any university or other institute of tertiary education. Information derived from the published or unpublished work of others has been acknowledged in the text, and a list of references is given.

Abstract

Student well-being concerns are increasing while educational support systems remain largely reactive. At the same time, Artificial Intelligence (AI)-mediated conversations are becoming a routine part of learning through educational assistants such as the European Union (EU)-funded AI4EDU project's Study Buddy. However, research in AI-based student well-being monitoring remains limited, with only one of 59 reviewed studies utilizing real interaction data, while 65% included no privacy component and 72% did not explicitly address ethical considerations.

This thesis develops and evaluates a privacy-first, human-reviewed well-being monitoring framework for educational chatbot conversations. The work is organized around three research questions: (RQ1) the feasibility of detecting well-being and engagement signals in educational conversations, evaluated zero-shot against a 54-session human-labeled gold set from the AI4EDU Study Buddy corpus; (RQ2) whether lightweight and multi-task transformer models provide sufficient performance for practical deployment; and (RQ3) whether privacy-by-design techniques – anonymization, context minimization, and on-premise inference – can be applied without degrading detection performance, scored against a published ethics-assessment framework for student-mental-health Machine Learning (ML) research.

The results show that educational conversational data present challenges not addressed by conventional sentence-level architectures: improved context modeling enhances detection, though zero-shot transfer to real educational conversations remains partial for theme and stress, and at chance level for risk. A comparative benchmark demonstrates that a rigorous evaluation protocol reduces the same RoBERTa-Large architecture's previously reported 93% macro-F1 (97% accuracy) result on risk classification to 0.81 macro-F1 on the same dataset – a genuine 12-point collapse on the same metric, with every other benchmarked model in the same 12–16 point range. DistilRoBERTa retains 97.6% and 96.3% of RoBERTa-Large performance on risk and thematic classification tasks, respectively, while achieving 6.7× lower CPU latency. Privacy-preserving approaches incur only minor utility loss, with anonymization reducing per-

formance by 0-1.8 macro-F1 points. Furthermore, an on-premise fine-tuned DistilRoBERTa model outperforms all evaluated cloud-based large language model variants on the highest-stakes risk-classification task by approximately 20 macro-F1 points while operating at roughly one-tenth of the latency. s The principal contribution of this work is a deployable, privacy-aware, and human-centered well-being monitoring framework that combines conversational analysis, temporal trend monitoring through the Mental-Health Compass, and structured human review. Rather than replacing educators, the proposed system strengthens existing student-support workflows by providing early awareness of potential well-being concerns while ensuring that all intervention decisions remain under human responsibility and control.

Acknowledgements

We would like to thank our thesis coordinator, Saleha Haseeb (Luleå University of Technology), for giving us the opportunity to work on this thesis and for her guidance and direction throughout — from the earliest seed papers in the literature review through the discussion of data strategy to pushing the scope of the Responsible AI questions from aspirational to measurable. This thesis would not have been possible without her support. We are also grateful to Abdullah Minhas for his support with the project's infrastructure and his guidance throughout the thesis; the *Mental-Health Compass* design draws inspiration, ideas, and tooling from his own Q&A portal, https://q2a.ai4pedagogy.ai/, which informed the deployed system at https://compass.ai4pedagogy.ai/. We also thank the organizers of the eRisk lab at the Conference and Labs of the Evaluation Forum (CLEF) — Javier Parapar, Anxo Pérez, Xihui Wang, and Fabio Crestani — for granting access to the eRisk 2025 Task 2 dataset under their signed user agreement, which supported this thesis's supplementary stress fine-tuning experiments.

Our sincere thanks to Magnus Selberg, school director at Nya Läroverket (NLV), Luleå, for early discussions on student well-being topics and how NLV conducts its surveys, for patiently listening to our thesis and *Mental-Health Compass* design, and for being consistently responsive and open — both in helping us with data collection/surveys and in discussing ideas for the platform's future development.

We also thank Camilla Grane, Senior Lecturer in Psychology at Luleå University of Technology, for her insights on psychological aspects of student well-being, and general direction on these topics.

This work is conducted in the framework of the project "AI-ready education: Mapping and innovating Generative AI in teaching and learning" (Project 101259658 – GenAI-EduMap – ERASMUS-EDU2025-PI-FORWARD), https://genai-edumap.eu/. Funded by the European Union. Views and opinions expressed are, however, those of the author(s) only and do not necessarily reflect those of the European Union or the European Education and Culture

Executive Agency (EACEA). Neither the European Union nor EACEA can be held responsible for them. This research is also financially supported by the European Regional Development Fund and the MARTINA project (no. 20367152).

Disclaimer

In the course of this thesis, a range of digital tools and services was employed to support different aspects of the research, development, documentation, and presentation of the work. Google Scholar was used to identify and access relevant academic literature and supporting background material, while Google Search was used to facilitate the exploration and identification of potential datasets. Claude Code was used as a supplementary tool for code debugging, troubleshooting, and selected aspects of the implementation of the *Mental-Health Compass*.· Overleaf was used for the preparation and typesetting of the thesis, Weights & Biases (W&B) was used for experiment tracking and logging, and draw.io was used to design diagrams and other visual representations included in the thesis.

The use of AI-based and generative tools was conducted with due consideration of their inherent limitations, including the possibility of producing inaccurate, incomplete, or misleading information. Such tools were therefore used as supplementary resources and not as substitutes for the author's independent analysis, reasoning, or academic judgment. Information, recommendations, and code modifications generated or supported by AI tools were critically reviewed by the author and, where appropriate, verified against relevant sources, documentation, or experimental results.

The author retains full responsibility for the accuracy, integrity, interpretation, and final presentation of the work, including all methodological decisions, implementations, analyses, and conclusions presented in this thesis.

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Acronyms

Acronym	Meaning
AI	Artificial Intelligence
AI4EDU	AI for Education (Study Buddy corpus dataset)
API	Application Programming Interface
AUC	Area Under the Curve
BERT	Bidirectional Encoder Representations from Transformers
CC-BY	Creative Commons Attribution (license)
CLEF	Conference and Labs of the Evaluation Forum
CPU	Central Processing Unit
CSV	Comma-Separated Values
ECE	Expected Calibration Error
EU	European Union
GDPR	General Data Protection Regulation
GLSEN	Gay, Lesbian & Straight Education Network
GLUE	General Language Understanding Evaluation
GPT	Generative Pre-trained Transformer
GPU	Graphics Processing Unit
HITL	Human-in-the-Loop
HTTP	HyperText Transfer Protocol
IIoT	Industrial Internet of Things
INT8	8-bit Integer (quantization)
JSON	JavaScript Object Notation
KPI	Key Performance Indicator

Acronym	Meaning
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
LSH	Locality-Sensitive Hashing
MB	Megabyte
MH	Mental Health
MIL	Multiple Instance Learning
ML	Machine Learning
MTL	Multi-Task Learning
NLP	Natural Language Processing
NLV	Nya Läroverket (nlv.se)
ODC-PDDL	Open Data Commons Public Domain Dedication and License
ONNX	Open Neural Network Exchange
PII	Personally Identifiable Information
PISA	Programme for International Student Assessment
PRAW	Python Reddit API Wrapper
RAG	Retrieval-Augmented Generation
ROC	Receiver Operating Characteristic
SAD	Stress Annotated Dataset
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
ToS	Terms of Service
WHO	World Health Organization
XAI	Explainable Artificial Intelligence
YRBSS	Youth Risk Behavior Surveillance System

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Chapter 1

Introduction

Student well-being has emerged as a critical priority for educational institutions, as learners increasingly face academic pressure, emotional challenges, stress, anxiety, and engagement-related difficulties that can affect both their learning experience and overall development. At the same time, digital learning environments and Artificial Intelligence (AI)-powered educational platforms are becoming an integral part of modern education, creating new forms of interaction between students and intelligent systems. These interactions generate a rich source of conversational data that may provide valuable insights into students' well-being, learning experiences, and support needs.

Recent advances in Natural Language Processing (NLP), transformer architectures, and Large Language Models (LLMs) have demonstrated strong capabilities in understanding human language, sentiment, emotions, and thematic patterns within conversations. While these technologies have been widely explored in domains such as social media and mental-health-related text analysis, their application within authentic educational conversation data remains limited. Furthermore, existing research often prioritizes predictive performance while giving comparatively less attention to privacy, ethics, transparency, and responsible deployment in educational settings.

Consequently, a gap exists between the growing capabilities of AI-based conversational analysis and its practical, responsible adoption in education. Current approaches rarely analyze real student-chatbot interactions, generally focus on isolated messages rather than behavioral trends over time, and often lack clear mechanisms for human oversight and integration into existing student-support processes. As a result, educational institutions have limited AI-assisted tools that can help them identify emerging well-being concerns while maintaining ethical and peda-

gogical responsibility.

This thesis addresses these challenges by developing a privacy-aware, AI-augmented framework for student well-being monitoring through conversational analysis. The proposed approach combines NLP techniques, LLM-assisted thematic analysis, temporal trend monitoring through the Mental-Health Compass, and human-in-the-loop review mechanisms to identify meaningful patterns within educational conversations. Importantly, the system is not intended to replace teachers, mentors, counselors, or student-support teams; rather, it functions as a decision-support tool that assists them in recognizing potential concerns, prioritizing attention, and making more informed decisions while ensuring that all interventions remain under human judgement and control.

Decision-support framing, precisely: the system this thesis builds produces *flags for human review*, never assessments of students. Its output is an input to an existing school process (e.g., the Swedish *elevhälsa* team meeting) in which trained staff decide — the same pattern as attendance-based early-warning systems already standard in schools, applied to what students write instead of whether they show up.

1.1 Motivation

Educational institutions increasingly rely on digital learning platforms and AI-powered assistants to support students throughout their learning journey. These systems create continuous streams of conversational data that capture student interactions in a natural educational context as in AI4EDU project [Vacalopoulou et al., 2024, Monteiro and Mokayed, 2024]. Unlike traditional indicators such as grades, attendance records, or periodic surveys, conversational data has the potential to reveal changes in engagement, motivation, and well-being as they emerge over time. Consequently, it offers a new opportunity to support earlier identification of students who may benefit from additional attention or support.

Despite this potential, relatively little research has investigated well-being monitoring using authentic educational conversations. Most existing approaches are developed on social media or survey-based data and are rarely evaluated for their suitability in educational settings, where interactions are typically shorter, task-oriented, and pedagogically structured. Furthermore, aspects such as privacy preservation, human oversight, and deployment feasibility are often treated as secondary concerns rather than integral design requirements. This thesis is therefore

motivated by the need for a practical and responsible framework that combines conversational analysis with privacy-by-design principles, temporal trend monitoring, and human-in-the-loop decision support, enabling AI to assist educators while keeping all intervention decisions under human control.

As AI-mediated student conversation is a new, high-volume, largely unmonitored signal channel; the NLP field has the technical capability to analyze it (transformers, LLMs) but has not yet applied that capability responsibly to this modality, nor demonstrated it works on genuine educational conversation rather than social-media proxies. This thesis treats Responsible AI (privacy, ethics, human oversight) as an evaluated design dimension rather than a compliance afterthought, because the literature review (Chapter 2) shows this is exactly where the field is weakest.

1.2 Problem Statement

This thesis is organized around three research questions, restated here as evaluable problem statements:

RQ1 How effectively can NLP models and LLMs identify early indicators of student well-being and academic-engagement risk in educational chatbot conversations, including real educational conversations (the AI4EDU Study Buddy corpus) where well-being signals may be rare?

RQ2 Does a multi-task model (joint theme detection, stress detection and risk classification) outperform single-task models of the same capacity, and does a lightweight model (DistilBERT-class) retain enough of a larger model's performance to justify on-premise deployment?

RQ3 To what extent can privacy-by-design techniques — anonymization before model input, context minimization, and on-premise lightweight inference — be applied to a student well-being pipeline without degrading detection performance, and does the resulting system meet the published ethics criteria for student-mental-health Machine Learning (ML) research [Drira et al., 2024]?

Chapter 3 details how each question is operationalized; Chapter 4 reports the evidence for all three research questions, including RQ1's zero-shot transfer evaluation against a human-labeled

gold set.

1.2.1 Scope

This thesis focuses on the design and evaluation of a responsible AI framework for monitoring student well-being through conversational analysis in educational chatbot environments. The research investigates how educational conversations can be analyzed to identify well-being related themes and academic disengagement signals while preserving privacy and maintaining human oversight.

In Scope

- NLP analysis of educational conversational data.
- Detection of well-being-related themes and academic-disengagement indicators.
- Temporal aggregation and trend-based monitoring through the Mental-Health Compass.
- Comparative benchmarking of machine learning and transformer-based models, including efficiency and latency evaluation.
- Privacy-by-design mechanisms, including anonymization and context minimization.
- Human-in-the-loop workflows to support educator review and decision-making.

Out of Scope

- Clinical diagnosis, psychological assessment, or severity grading.
- Fully automated intervention without human review.
- Post-hoc explainability (XAI); interpretability is achieved through thematic labels, confidence scores, and evidence-based review mechanisms. Notably, only 2 of 59 reviewed studies incorporated XAI techniques [Drira et al., 2024].
- Large-scale real-world institutional deployment.
- Suicide or self-harm detection as a validated system capability.
- Multimodal well-being monitoring using audio, video, biometric, or behavioural data; this work is limited to text-based conversational analysis.

Scope Boundary: The proposed system is intended as a decision-support tool, not a diagnostic system. Its role is to assist teachers, mentors, counselors, and student-support teams in identifying potential well-being concerns, while all interpretation and intervention decisions remain under human responsibility and control.

1.3 Equality and Ethics

The proposed system is explicitly designed as a non-diagnostic decision-support tool and never acts autonomously. Any well-being-related flag generated by the system is reviewed by an educator, counselor, or support staff member before any action is taken, ensuring that all intervention decisions remain under Human-in-the-Loop (HITL) oversight.

No new student data was collected for this research. All experiments were conducted using publicly available or previously consented datasets. The AI4EDU Study Buddy [Vacalopoulou et al., 2024] corpus is used exclusively for zero-shot evaluation (RQ1) and is never included in model training, reducing the risks associated with the secondary use of educational data.

Privacy-preserving measures, including anonymization and context minimization, are incorporated into the proposed framework. Ethical considerations are evaluated using the student mental-health ML ethics framework of Drira et al. [2024], covering privacy, transparency, replicability, interpretability, and fairness, thereby treating Responsible AI as a core evaluation criterion rather than an afterthought in deployment; the fairness score, and the demographic gap behind it, is discussed in Section 4.3.4.

1.4 Organization of the Thesis

Chapter 2 reviews the literature and derives five priority-ordered gaps that motivate this thesis. Chapter 3 describes the data assets, the modular pipeline (classifier, *Mental-Health Compass* temporal aggregation, escalation, evidence-quoting reviewer summary), the comparative benchmark protocol (RQ2), and the privacy-ablation and ethics-audit design (RQ3). Chapter 4 reports the benchmark and privacy-arm results obtained to date. Chapter 5 concludes and discusses limitations and future work. Code and data availability is addressed in Section A.8.

Chapter 2

Literature Review

2.1 Overall Findings from Literature

This chapter reviews recent research on AI-based well-being monitoring, mental-health text analysis, and educational NLP. The objective is to examine the current state of the field, identify methodological limitations, and establish the research gaps that motivate this thesis. Particular attention is given to transformer-based architectures, LLMs, Responsible AI considerations, and deployment-oriented research relevant to educational environments.

The reviewed studies show that transformer models and LLMs substantially outperform classical machine-learning approaches on mental-health related text analysis tasks. LLM-based topic assignment reached Purity 0.75 / NMI 0.67 against subreddit ground truth, compared with 0.23 / 0.12 for Latent Dirichlet Allocation (LDA) and 0.45 / 0.40 for BERTopic [Zhao and Chen, 2025]. Fine-tuned RoBERTa-Large reports 94% F1 (93% macro-F1) on 7-class mental-health classification [Khan et al., 2025]; while LLM-assisted thematic analysis reduced qualitative coding effort from approximately 110 to 40 person-hours while recovering most human-identified codes [Parkington et al., 2025].

However, several studies caution against assuming that larger or more generative models necessarily produce superior outcomes. A fine-tuned BERT encoder (F1 0.78) achieved performance comparable to GPT-3.5 (0.80) on student-feedback sentiment analysis [Sood et al., 2024]. Similarly, adding retrieval-augmented generation (RAG) to GPT-4o reduced excerpt faithfulness by producing interpretive summaries instead of verbatim evidence [Parkington et al., 2025]. Reproducibility is also uneven: the RoBERTa-Large result above, for instance, does not fully specify the experimental setup used to obtain it [Khan et al., 2025]. These findings suggest

that model effectiveness depends not only on model scale but also on task design, grounding strategy, and deployment context.

Collectively, the reviewed studies demonstrate substantial progress in AI-based mental-health text analysis. Nevertheless, important limitations remain regarding educational applicability, privacy protection, temporal reasoning, human oversight, and deployment readiness. These limitations motivate the five research gaps identified below.

Gap 1 — No Educational Conversational Data

None of the reviewed studies analyzes student–chatbot conversations or educational conversational data. The corpora used are Reddit/Twitter posts [Zhao and Chen, 2025, Khan et al., 2025], single-turn survey feedback [Sood et al., 2024], and healthcare-worker interviews [Parkington et al., 2025]. An ethics-centered review of 59 student mental-health ML studies found exactly one study using human-agent interaction data, with 50% relying on bespoke surveys [Drira et al., 2024]. A broader LLM mental-health scoping review (29 studies) similarly reported social media as the dominant data source, with no educational conversational data represented [Yang et al., 2026].

Gap 2 — Privacy and Ethics Are Recommended, Not Implemented

Privacy and ethical considerations remain largely aspirational within the literature. Drira et al. [Drira et al., 2024] found that 65% of reviewed studies included no privacy component, while 72% did not explicitly mention ethics. Although privacy-preserving techniques such as differential privacy, federated learning, and synthetic data generation are frequently recommended, none of the surveyed studies implemented these approaches in practice.

Gap 3 — Absence of Temporal and Multi-Task well-being Modelling

Most existing systems classify messages in isolation and operate as single-task predictors. No reviewed study aggregates well-being indicators over time to identify behavioural trends or persistent well-being concerns. Similarly, longitudinal signal analysis and integrated multi-task frameworks remain largely unexplored. A broader mental-health scoping review independently identifies modelling longitudinal user history as an open challenge [Yang et al., 2026].

Gap 4 — Human Oversight Is Endorsed but Not Designed

Human involvement is commonly recommended but rarely operationalized. Existing approaches incorporate humans primarily through annotation [Sood et al., 2024], post-hoc inspection [Zhao and Chen, 2025], or recommended hybrid workflows [Parkington et al., 2025]. None of the reviewed systems places a human reviewer directly at the intervention decision point through a defined escalation and review protocol. Furthermore, hallucinated clinical information has been identified as a critical risk for AI-supported mental-health systems [Yang et al., 2026].

Gap 5 — Deployment Efficiency Is Asserted, Not Measured

Although lightweight transformer models are well motivated within the broader NLP literature, deployment-oriented evaluation remains limited. DistilBERT demonstrated a 40% parameter reduction and approximately 60% lower latency while retaining 97% of BERT performance on General Language Understanding Evaluation (GLUE) benchmarks, a suite of general-English sentence-understanding tasks [Sanh et al., 2019]. However, none of the reviewed mental-health studies systematically evaluates latency, memory footprint, calibration, or deployment cost, and no DistilBERT-class model had previously been evaluated on educational conversational data.

These five gaps collectively define the motivation and contribution of this thesis: a privacy-first (Gap 2), temporally aggregating (Gap 3), human-reviewed (Gap 4) well-being decision-support framework, evaluated using authentic educational chatbot conversations (Gap 1) and supported by measured performance–efficiency trade-offs across model classes (Gap 5). Related evaluation work on the same AI4EDU platform has examined tool-adoption outcomes [Haseeb et al., 2026]; this thesis instead evaluates well-being and risk-detection performance on the educational conversational data.

2.2 Model Landscape

Table 2.1 summarizes the model classes included in the comparative benchmark, while Table 2.2 presents verified head-to-head comparisons from the literature.

Table 2.1: Model classes in the comparative benchmark.

Model	Params	Origin
TF-IDF + SVM/LogReg	<1M	field default (33% of studies, Drira et al., 2024)
DistilBERT	66M	6-layer distilled BERT [Sanh et al., 2019]
DistilRoBERTa	82M	6-layer distilled RoBERTa, same recipe as Sanh et al. [2019]
BERT-base	110M	Devlin et al. [2019]
MentalBERT	110M	BERT + 13.7M mental-health Reddit sentences [Ji et al., 2022]
RoBERTa-base	125M	Liu et al. [2019]
Mental-RoBERTa	125M	RoBERTa + mental-health pretraining [Ji et al., 2022]
RoBERTa-Large	355M	the Frontiers paper's model [Khan et al., 2025]

TF-IDF + SVM/LogReg is field context only, not run in this thesis's own benchmark (Table 4.5).

Table 2.2: Verified head-to-head model comparisons (mental-health and general text).

Study	Data	Key result	DistilBERT retention
Sanh et al. [2019]	GLUE/IMDb	DistilBERT 77.0 vs BERT 79.5 GLUE	97% of BERT
Adoma et al. [2020]	ISEAR emotions	RoBERTa .743 > BERT .701 > DistilBERT .669	90% of RoBERTa (worst case)
Cortiz [2022]	GoEmotions (58k comments)	DistilBERT .48 > BERT .46; RoBERTa .49	104% of BERT, 98% of RoBERTa
Galke et al. [2022] (12 datasets)	mixed	DistilBERT wins GoEmotions outright	~99% of BERT avg
Hussain et al. [2025]†	Reddit, 4 depression levels	MentalBERT 97.3 > RoBERTa 96.3 > BERT 95.2 > DistilBERT 95.1 F1	unreliable, see note
Aggarwal et al. [2024]	9 disorders, weak labels	DistilBERT 86.3% beats MentalRoBERTa 82.7%	≥100% under weak labels
Hasan et al. [2025]	120k Reddit posts	RoBERTa 95.96 > DistilBERT 94.97 ext-F1	99% of RoBERTa
Berghia and Barglazan [2025] (minor venue)	suicide risk binary	RoBERTa cut false negatives 40% vs DistilBERT	caution → tiered design

†Downgraded after full review — single run, private dataset, unspecified F1 averaging; cited only as a negative methodology example.

The literature indicates that lightweight transformer models can retain a substantial proportion of the performance of much larger architectures while significantly reducing computational requirements. Several studies report DistilBERT-class models achieving approximately 97%–99% of the performance of larger transformer models, particularly in short-text and conversational classification tasks Sanh et al. [2019], Cortiz [2022], Galke et al. [2022]. This efficiency is especially relevant for practical deployment, where model quality must be considered together with latency, memory footprint, calibration, and operational cost. However, these deployment-oriented dimensions remain comparatively underexplored in prior work.

The importance of this broader system-level perspective is not unique to modern language models. Across applied AI, model development has progressively shifted from highly task-specific pipelines, in which performance depended heavily on handcrafted features, explicit thresholds, and manually designed decision mechanisms, toward architectures that learn increasingly rich representations directly from data. Earlier applied systems illustrate this paradigm clearly, including edge-based detection pipelines for license-plate localization [Mokayed et al.,

2014] and robust thresholding strategies for structured-document classification [Mokayed and Mohamed, 2014]. In such systems, successful deployment depended not only on the choice of classifier, but also on how well the complete processing pipeline was adapted to the characteristics of the target data and operating environment.

Although contemporary deep-learning systems reduce the dependence on handcrafted representations, the underlying deployment challenge remains. Application-specific studies continue to show that models developed for one setting cannot be judged solely by benchmark accuracy, since changes in data distribution, sensing conditions, computational resources, and operational constraints can substantially affect their usefulness in practice [Mokayed et al., 2025]. This is particularly important for educational conversational AI, where deployment introduces additional requirements beyond predictive accuracy, including computational efficiency, calibration, privacy, responsiveness, and suitability for local or on-premise execution.

Accordingly, the model comparison in this thesis is framed not simply as a search for the highest-performing architecture, but as an evaluation of the trade-off between predictive effectiveness and practical deployability. This motivates the comparison of lightweight and larger transformer models using not only macro-F1, but also latency, calibration, model size, and deployment-related considerations.

2.3 Positioning

The five identified gaps directly motivate the three research questions presented in Chapter 1 and define the contributions of this thesis.

- **Applied Contribution:** A well-being decision-support framework designed specifically for educational chatbot conversations, addressing the lack of educational conversational data and temporal monitoring mechanisms (Gaps 1, 3, 4; RQ1).
- **Scientific Contribution:** A rigorous comparative benchmark evaluating effectiveness, calibration, efficiency, and deployment trade-offs across classical, transformer-based, and domain-adapted models (Gap 5; RQ2).
- **Responsible AI Contribution:** A privacy-by-design framework incorporating ethical evaluation, human oversight, and deployment considerations within educational well-being monitoring (Gaps 2, 4; RQ3).

Three deliberate scope decisions further define this positioning:

- **Explainability** is excluded as a primary research gap, since only two of the 59 studies reviewed by Drira et al. [Drira et al., 2024] implement any form of XAI.
- **Multi-task Learning (MTL)** is treated as an empirical question rather than a presumed benefit and is therefore evaluated through targeted ablation studies (RQ2).
- **Full RAG** is intentionally replaced by an evidence-quoting reviewer summary mechanism (Section 3.3.4), motivated by findings that RAG can reduce excerpt faithfulness in mental-health thematic analysis [Parkington et al., 2025].

By addressing educational applicability, temporal analysis, privacy preservation, human oversight, and deployment feasibility within a single framework, this thesis seeks to bridge the gap between advances in conversational analysis and the practical requirements of responsible educational deployment [Karafyllidis et al., 2025].

Chapter 3

Methodology

3.1 Data Understanding

3.1.1 Data Strategy

No public dataset matches this thesis's target modality — genuine student–chatbot conversation annotated for well-being signal — so the data strategy is deliberately scoped into named tracks, each with one job, rather than attempting a single do-everything corpus:

- **Track A (training, in scope).** Public proxy datasets (SAD, Kaggle MH aggregate) provide the labeled volume needed to train and benchmark classifiers. Domain shift from these datasets to real educational text is treated as an explicit, measured finding, not hidden.
- **Track B (test-only, in scope).** The AI4EDU Study Buddy corpus provides the one real educational-conversation evaluation this thesis has, used exclusively zero-shot to avoid weak-label circularity; how its gold labels and weak labels are produced is detailed in Section 3.2.
- **Track C (eRisk supplementary fine-tuning) — in scope, completed.** Access to the eRisk 2025 Task 2 dataset [Losada and Crestani, 2016, Parapar et al., 2025] (909 subjects, real threaded depression-forum text with ground-truth labels) was obtained after the original access request; used to fine-tune the stress head (depression label proxied as is_stressor, an assumption stated and not validated as equivalence). eRisk is a curated shared-task release with a fixed train/test split and genuine ground-truth labels – distinct

from the unlabelled Reddit corpus used for Compass demonstration (below).. The headline, robust result is a false-positive-rate reduction of ~85–87% on 329 real, mostly-benign AI4EDU deployment-like sessions (the 383-session Ireland corpus minus the 54-session goldset, Table 3.1; versus the fragile n=1-positive AI4EDU gold-setdelta), alongside a genuine in-domain aggregate-detection gain confirmed not to be a verbosity/volume shortcut (a held-out above-median word-count ablation scored higher, not lower, ruling out that confound). This does not establish improved true-positive stress detection on AI4EDU itself, which remains open pending more gold labels (Section 4.1).

- **Track D (synthetic distress data) — scoped out.** LLM-generated, psychologist-reviewed synthetic student–chatbot dialogue was considered as a stress-test corpus, but synthetic dialogue's validity as a proxy for genuine distress language is itself questionable, not merely a fallback of last resort. Once eRisk (Track C) and AI4EDU (Track B) provided real data for the roles synthetic dialogue would have filled, the fallback became unnecessary.

- **Track E (short-text survey signal) — pending, not scoped out.** Sweden's NLV school survey data was investigated as a lightweight, short-text well-being-signal precedent, fitting the *Mental-Health Compass*'s periodic survey input path by design (Table 3.4). The sample obtained so far is too small and free-text-sparse to support any classifier role in this thesis — a data-volume limitation, not a decision against the modality.

The Greek-language subset of AI4EDU (Table 3.1) was explicitly decided against rather than left open; the NLV survey remains genuinely pending on sample size rather than decided against (Section 3.1.1) — each candidate data source is given one clear status (used, decided against, or pending) rather than an undifferentiated "future work." Table 3.1 allocates every dataset that *is* used in this thesis to exactly one job, so that every claim maps to data that can actually support it.

Table 3.1: Data-role architecture — one job per dataset.

Role	Data	Rationale
Train classifiers (theme/stress: SAD; risk: Kaggle)	SAD [Mauriello et al., 2021] (theme + stress) + Kaggle MH aggregate [Sarkar, 2024] (risk)	labeled, sufficient volume; SAD's short SMS-like sentences are the closest public proxy to chat turns
Test educational transfer (Gap 1)	AI4EDU Study Buddy corpus [Stamouli and Paraskevopoulos, 2026] — English-language Ireland sessions only (383 of 601), held-out, zero-shot, test-only, never training	no labels, rare positives, short curricular turns; also avoids weak-label circularity; Greek-language Greece/Cyprus sessions (218 of 601) excluded — English-only PII-masking and classifier stack, a deliberate scoping decision
Supplementary stress fine-tuning (Track C)	eRisk 2025 Task 2 [Losada and Crestani, 2016, Parapar et al., 2025] (909 subjects, real threaded depression-forum text, ground-truth labels)	real, age-relevant supplementary data unavailable elsewhere; depression label proxied as is_stressor (stated assumption, not validated equivalence); yields a robust false-positive-rate reduction on real AI4EDU-like text, not a validated AI4EDU stress-detection improvement
Demonstrate trends (*Mental-Health Compass*)	timestamped public Reddit data [Low et al., 2020] (577,033 posts, 13 subreddits selected for theme/risk taxonomy relevance), reshaped by per-author *stitching* into a chat-like pseudo-session corpus (see Section 3.2)	AI4EDU is statistically underpowered for trends (117 students, ~5 sessions each, rare signals); Reddit substitutes as a face-validity testbed for the eRisk-fine-tuned checkpoint (Track C), which substantially improves severity-discrimination over the plain zero-shot classifier – full evidence in Section A.7
Deployment modes (design context)	NLV school survey records (Sweden, structured smiley/Likert + free text)	schools currently have little chat; surveys naturally produce per-student timestamped sequences the *Mental-Health Compass* needs

Label taxonomy and its well-being grounding. The classifier's labels are inherited from its training data rather than derived from a psychological model: nine theme classes from SAD [Mauriello et al., 2021] (Emotional Turmoil, Everyday Decision Making, Family Issues, Financial Problem, Health/Fatigue/Physical Pain, School, Social Relationships, Work, and Other), a binary stress label, and seven risk categories from the Kaggle aggregate(Anxiety, Bipolar,Depression, Normal, Personality disorder, Stress, Suicidal). Labelled data determines what can be trained, but the taxonomy can still be read against established theory: in the transactional model of stress [Lazarus and Folkman, 1984], stress arises when a demand is appraised as exceeding one's resources, so the theme head approximates the stressor domain while the stress head approximates the appraised strain, and the *Mental-Health Compass* persistence rule (Section 3.3.2) echoes the classic chronic/acute stressor distinction by separating recurring from one-off concerns. The themes can also be loosely read against the three needs in self-determination theory [Ryan and Deci, 2000] (e.g. School/Work as competence, Family Issues/Social Relationships as relatedness), though this mapping is illustrative rather than a validated construct. Aligning the taxonomy with the categories used by Swedish student-health (*elevhälsa*) teams is left to future work with practitioners.

3.1.2 Additional Data Sources Explored

For each of three threads — conversational, survey, and Reddit teen-identified — a systematic search combined age terms (high-school, adolescent, teen), modality terms (chatbot, conversation, survey, free-text), and outcome terms (mental health, well-being, depression, anxiety), run across PubMed and general web/Scholar-indexed search (e.g., *"adolescent chatbot conversation text dataset depression anxiety"*, *"r/teenagers Reddit dataset mental health posts age-verified download"*). Every candidate surfaced was checked against three criteria — concrete downloadability, verified age-relevance, and documented provenance — rather than accepted on abstract text alone. The search was independently cross-validated against a 2025 scoping review cataloging public youth-mental-health datasets [Min et al., 2025], which found no dataset meeting these criteria beyond what this search had already identified — corroborating, not merely asserting, that Gap 1 remains open. Full search trail in Section A.1

3.2 Data Preparation

The data layer ingests public proxies (Kaggle MH aggregate, SAD) plus the AI4EDU Zenodo corpus (test-only, per Table 3.1), with deduplication (Section 3.3.1) and leakage controls and a pseudonymization-at-ingestion module that also serves as the RQ3.1 privacy-ablation switch. The eRisk 2025 Task 2 supplementary fine-tuning data (Track C) is not deduplicated by the same protocol – a curated shared-task release rather than a scraped/merged aggregate. A cross-boundary MinHash-LSH check (Jaccard≥ 0.9, same protocol as the SAD/Kaggle dedup) found zero genuine content-leakage between the train and holdout splits; all flagged matches were dataset placeholders or generic reused phrases.

The Reddit corpus used for Compass trend demonstration (Table 3.1) is prepared differently, by *stitching*: each qualifying author's own posts (≥5 posts spanning ≥28 days) are ordered chronologically into a pseudo-session, with long posts chunked to ≤280 characters to resemble AI4EDU's short-turn register. This is not real dialogue – one author posting over time, not two parties turn-taking – and, unlike eRisk, it has no ground-truth labels and is never used for training or evaluation; its sole role is the face-validity check reported in Section A.7.

AI4EDU arrives with no theme/stress/risk labels at all, and the classes that matter most are extremely rare (as few as 1–2 positive sessions out of 383) – random sampling for human annotation would likely miss them entirely. A weak-labeling layer (a controlled-prompt LLM, Qwen2.5-32B-Instruct, selected over Llama-3.3-70B in a bake-off despite being 2× smaller, Section A.7) produces a candidate theme/stress/risk label with a confidence score for every session, used solely to *select* which sessions go to human annotation, stratified across every theme class and both stress-positive/risk-flagged sessions, supplemented with a random sample for a combined 54-session gold set (Section A.7.1). Ground truth for evaluation is exclusively the resulting human-validated gold subset; weak labels themselves are never used as training data or evaluation ground truth, which avoids the labeler grading itself. Every message that reaches a model – whether in this thesis's own evaluation pipeline or the live student-facing ingestion service – passes through a shared Personally Identifiable Information (PII)-masking HyperText Transfer Protocol (HTTP) service (Presidio + spaCy en_core_web_sm), fail-closed on service-down (Section 3.3.5 details the exact behavior).

3.3 Modelling

3.3.1 Comparative Benchmark (RQ2)

The benchmark grid spans DistilBERT/DistilRoBERTa → BERT-base/MentalBERT → RoBERTa-base/Mental-RoBERTa → RoBERTa-Large → LLM zero/few-shot (Table 2.1), each in single- and multi-task variants (joint theme + stress + risk heads), on two public datasets: SAD (6,824 sentences after deduplication; theme and binary stress tasks) and the Kaggle mental-health aggregate (50,740 statements after deduplication; 7-class risk task). The protocol adds everything the closest prior work [Khan et al., 2025] does not report: deduplication before splitting (Table 3.2), with a cross-boundary leakage check confirming no near-duplicates crossed train/test; stratified 70/15/15 manifests with a fixed published seed; an optional group-aware split by source dataset (a leakage probe); per-model hyperparameter tuning; three seeds with mean±std and McNemar/bootstrap confidence intervals; and, uniquely for this field, measured calibration (Expected Calibration Error, ECE; reliability diagrams), latency, and model size alongside macro-F1.

Deduplication protocol. Both training corpora are deduplicated in two stages, applied to the full corpus *before* any train/val/test split, so that no near-copy of a training example can leak into the test set. First, exact duplicates are removed by comparing normalized text (lowercased, non-alphanumeric characters stripped, whitespace collapsed), keeping the first occurrence. Second, near-duplicates are found with MinHash locality-sensitive hashing (LSH) over character 3-gram shingles (64 permutations, 16 bands; parameter rationale in Section A.2); any candidate pair surfaced by the LSH banding is then verified against the true Jaccard similarity of its shingle sets, and the later occurrence is dropped if similarity ≥ 0.9. The method follows established large-scale text-deduplication practice [Lee et al., 2022], applied here to a mental-health dataset for the first time in this literature. Table 3.2 reports the resulting counts for both corpora.

Table 3.2: Deduplication counts, both training corpora.

Dataset	Raw input	Exact removed	Near removed (≥ 0.9)	Final
SAD	6,850	21	5	6,824 (0.38%)
Kaggle MH aggregate	52,681	1,721	220	50,740 (3.68%)

Four pre-registered hypotheses structure RQ2:

- **H1** RoBERTa-Large scores well below the published 93% macro-F1 (97% accuracy) re-

ported by Khan et al. [2025] under a rigorous protocol.

- **H2** A DistilBERT-class model retains ≥95% of RoBERTa-Large's macro-F1 at markedly lower latency, a threshold set from Sanh et al. [2019]'s 97% GLUE retention.
- **H3** Mental*-pretrained models gain at most ≤3 macro-F1 over their general-domain counterparts, the margin reported by Ji et al. [2022].
- **H4** Group-aware (source-held-out) splitting drops performance noticeably, evidencing style leakage and calibrating expectations for the AI4EDU transfer evaluation.

Hyperparameter protocol. Per-model tuning is not a single shared configuration applied uniformly (the setup that would, per Galke et al. [2022], handicap learning-rate-sensitive models such as DistilBERT): for every model, a learning rate is selected on the validation split from the sweep $\{1, 2, 3, 5\} \times 10^{-5}$ before the three-seed test-split run at the selected rate — for example, the sweep selects 3×10^{-5} for DistilBERT versus 5×10^{-5} for DistilRoBERTa on theme/risk, confirming that a shared configuration would have penalized one of the two (full per-model, per-task selected rates in Table A.2). Training uses class-weighted cross-entropy and early stopping (patience 2) validation macro-F1, at epoch ceilings of 5 (theme/stress single-task) and 4 (risk/multi-task). To rule out under-training as a confound, every distil-class model/task cell was independently retrained at raised epoch ceilings (8 and 6 respectively, same early-stopping patience) as part of the reproducibility rerun below; macro-F1 deltas ranged from -0.013 to $+0.015$ across all 12 cells — the same magnitude as ordinary seed-to-seed noise, with no consistent direction — confirming the original ceilings were not a limiting factor for any reported result.

Reproducibility rerun. Beyond the three-seed variance already reported per cell, the full benchmark was independently re-executed end-to-end from the committed, fixed-seed split manifests on later dates, in two passes: the RQ3.1 privacy-ablation arms and H4's group-aware split (2026-07-20), and the five remaining benchmark architectures – BERT-base, MentalBERT, Mental-RoBERTa, RoBERTa-base, and RoBERTa-Large – across all four task configurations (2026-08-30). Every resulting (experiment, task, mode) cell was compared against the originally reported figure with an automated tolerance check. **51 of 53 cells** landed within ± 0.02 macro-F1 of the original run, including H1's headline result (RoBERTa-Large single-task risk:

original 0.8109, rerun 0.8136, delta +0.0027). The two cells outside tolerance are both accounted for: RoBERTa-Large single-task stress (−0.0556) corroborates, rather than contradicts, the large-model fine-tuning instability on small binary data already documented elsewhere in this thesis (per-seed collapse 0.721/0.640/0.484); BERT-base single-task stress (+0.0238) is a modest overshoot with no comparable instability history. This confirms the reported numbers are reproducible from the public repository and documented data-fetch scripts, not an artifact of one lucky run — the strongest-evidenced claim in the ethics-framework audit (Section 4.3.4).

3.3.2 The Mental-Health Compass

The *Mental-Health Compass* is a privacy-minimal, rolling temporal aggregate of well-being signals per pseudonymous student — a navigation instrument for educators, not a diagnosis record (Table 3.3). Classifier themes and confidence scores are aggregated over a rolling window (default seven days); the escalation layer reads *trends*, not single messages: a sentiment-slope rule (sustained decline across consecutive windows), a theme-persistence rule, a theme-novelty rule, and a theme-co-occurrence rule are all implemented and running, though none has yet been evaluated against labelled outcomes (window choices and red-flag combinations are configurable parameters, pending future survey or stakeholder data rather than tuned or validated at this stage). Figure 3.1 shows the aggregation mechanism end to end, from a per-student message stream to an escalation decision.

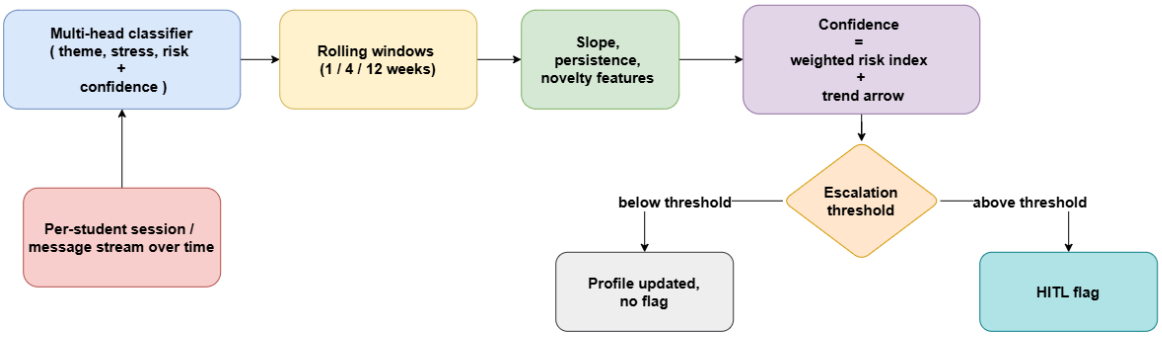


Figure 3.1: The *Mental-Health Compass* temporal-aggregation mechanism: classifier output is aggregated over rolling windows into trend features, which drive the escalation decision.

Privacy properties: the pseudonymization key is held by the institution; only aggregates are stored (never raw text); retention decays after N months; evidence fetch is access-controlled and audit-logged. No PII is present in the profile itself, making the *Mental-Health Compass* a concrete data-minimization instance, scoreable under the Drira privacy criteria (Section 4.3).

Table 3.3: Mental-Health Compass specification.

Field	Example	Deliberately *not* stored
Pseudonymous ID	stu_4f2a	name, contact, demographics
Theme trajectory	weekly presence rate per theme (stress, isolation, hopelessness)	raw conversations
Confidence-weighted risk index	rolling-window score	any clinical label
Flag history + reviewer outcomes	"2 flags: 1 confirmed, 1 dismissed"	reviewer free-text about the student
Evidence pointers	message IDs of flagged excerpts (fetched on demand, access-logged)	copies of excerpts

3.3.3 Escalation and Review Layer

A rule-based multi-level escalation layer reads the *Mental-Health Compass* against an explicit alert budget (flags per student-week, a configurable parameter pending future survey data) and routes flags into a human review queue; messages matching crisis patterns take a fast path to immediate human review, with maximum recall but no reliability claim (Section 1.2.1). Human-in-the-loop operates strictly at the *decision layer* : no action toward a student or educator is ever triggered by the model alone. A designated staff member confirms, dismisses, or escalates each flag; every decision is logged and reused as evaluation data (reviewer-agreement Key Performance Indicator, KPI; Section 3.5). This contrasts with the literature, where human involvement appears only at annotation time [Sood et al., 2024], as passive inspection [Zhao and Chen, 2025], or as a recommended hybrid workflow [Parkington et al., 2025] — none of the reviewed systems places a human at the intervention decision point. Doing so also aligns by design with the human-oversight requirements of the EU AI Act (Art. 14) [European Parliament and Council, 2024]. The Act's stricter rule for emotion inference in education (Art. 5(1)(f)) applies to biometric-based emotion recognition (per its own definition, Art. 3(39)); text-content analysis is not biometric, and the system's flags serve a safety purpose with a human always deciding.

3.3.4 Evidence-Quoting Reviewer Summary

The reviewer-facing summary is deliberately not a full RAG system [Monteiro and Mokayed, 2024]. It surfaces two evidence sources verbatim: (1) the student's own flagged messages, quoted with references — the reviewer verifies, never trusts a generated paraphrase; and (2) the

Mental-Health Compass trend numbers behind the flag (e.g., "rising 3 weeks"). A fuller RAG design (institutional support-protocol retrieval, LLM-generated citation-grounded summaries) was sketched but not built or evaluated (Section 3.3.5). This scoped-down choice is motivated by prior work finding RAG *reduced* excerpt fidelity in a comparable setting (29% vs 56% human-excerpt match; Parkington et al., 2025): quoting evidence directly sidesteps that failure mode rather than requiring it to be measured away.

3.3.5 Deployed System Architecture and Implementation Status

The *Mental-Health Compass* is a system, not a single model, and this thesis is explicit about which parts of that system are running today versus designed for later. The deployed pipeline is: raw student text → shared PII-masking service → multi-head DistilRoBERTa classifier (theme/stress/risk + confidence) → *Mental-Health Compass* rolling aggregation → escalation threshold → HITL review queue → logged reviewer decision (evaluation-only feedback, never fed back into training). The classifier itself is exported to Open Neural Network Exchange (ONNX) (fp32 and int8) with a validated release manifest and is integrated into the live student-facing analysis service as its multi-head DistilRoBERTa classifier. PII masking is one canonical implementation (Presidio + spaCy en_core_web_sm) with two callers: this project's own offline anonymization ablation (RQ3.1, in-process) and the live ingestion pipeline (over HTTP, deployed as a separate service), which fails closed — if the masking service is unreachable, ingestion of English text raises rather than silently storing unmasked text. Non-English messages are still stored unmasked, an explicit, documented limitation rather than a silent one.

Architecture. The classifier (**v2**) is DistilRoBERTa-base (82M parameters) with a shared encoder and three linear heads (theme, stress, risk). Pooling uses a learned attention mechanism — a linear scorer over token positions, masked-softmax-weighted sum of the encoder's last hidden state — rather than a single fixed token position, so the classifier can weight whichever tokens carry signal across a multi-turn exchange, at a negligible parameter cost (Section A.4). The input window is 512 tokens (DistilRoBERTa reserves 2 of its 514 positional embeddings for special tokens), sized to fit a full student–tutor exchange rather than a single short sentence. A learning-rate sweep (Section 3.3.1's protocol applied to the multi-task checkpoint) selects 5×10^{-5}. In-domain (SAD/Kaggle MH, same protocol as Section 3.3.1, held-out test split, mean ± sd of 3 seeds) macro-F1: theme $0.7117 \pm .0065$, stress $0.7027 \pm .0202$, risk $\mathbf{0.8100} \pm .0089$

— this is the classifier described everywhere else in this thesis (this configuration — attention pooling, 512 tokens — is deliberately distinct from the uniform CLS-pooling, 128-token protocol used for cross-model comparability in the RQ2 Stage-A grid, Table 4.5; the two are not the same run; release artifact details in Section A.4). Figure 3.2 shows this pipeline end to end. Whether the full 512-token window should carry the tutor's turns as well as the student's is evaluated separately in the zero-shot transfer analysis (Section 4.1). So is **v2-MIL** (built on Multiple Instance Learning, MIL) — a further, zero-parameter inference-time strategy that encodes a session as a bag of turns under v2's unchanged weights (Figure 4.2) — since both are inference-time choices, not changes to the trained weights, and are kept out of this architecture description accordingly.

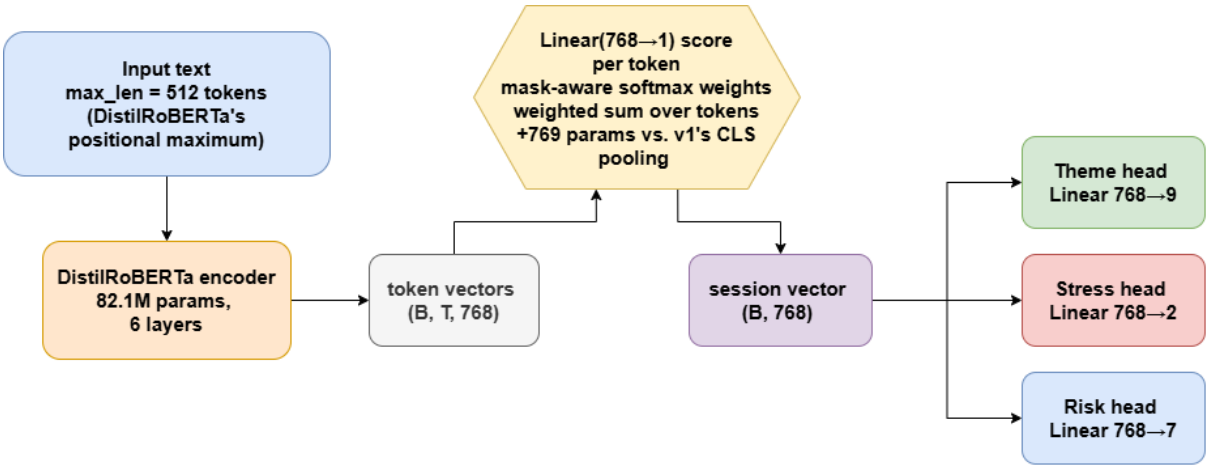


Figure 3.2: v2 architecture: attention pooling over the full 512-token encoder output, replacing v1's single-position CLS pooling.

The system is designed to accept signal from more than one input modality into the same per-student profile, but not all of them are built. Table 3.4 states this honestly, distinguishing implemented paths from designed-but-not-built ones, so the multi-modal claim in this thesis is read as a *design* claim (a source-agnostic taxonomy and aggregation mechanism, demonstrated on two of four intended input paths) rather than a claim that all four modalities run in production today.

Figure 3.3 shows the full deployed pipeline: the two training corpora feeding the shared encoder, the three implemented/designed input paths, the shared PII-masking service sitting between raw text and the model, and the flow from classifier output through *Mental-Health Compass* aggregation, escalation, and the human review queue — including the evaluation-only (never-training) feedback loop from logged reviewer decisions.

Table 3.4: Compass input-modality implementation status.

Input modality	Status	Notes
Chat conversations (AI4EDU)	**Implemented**	the path evaluated in this thesis; sessions stitched per-student, chronologically ordered
Social-media-proxy trend demo (Reddit)	**Implemented**	scale demonstration on public timestamped data, not a training signal (Table 3.1)
Periodic survey mode	Prototype adapter built, not production-integrated	a generic academic/survey Comma-Separated Values (CSV) ingestion adapter and analyzer exist as a demo/test-data path; no production survey-cadence integration
Educator/administrative signal (grades, attendance, teacher notes)	Designed, not built, not acquired	would bypass the text classifier and feed *Mental-Health Compass* aggregation directly as a correlated signal; AI4EDU itself contains no academic-performance records

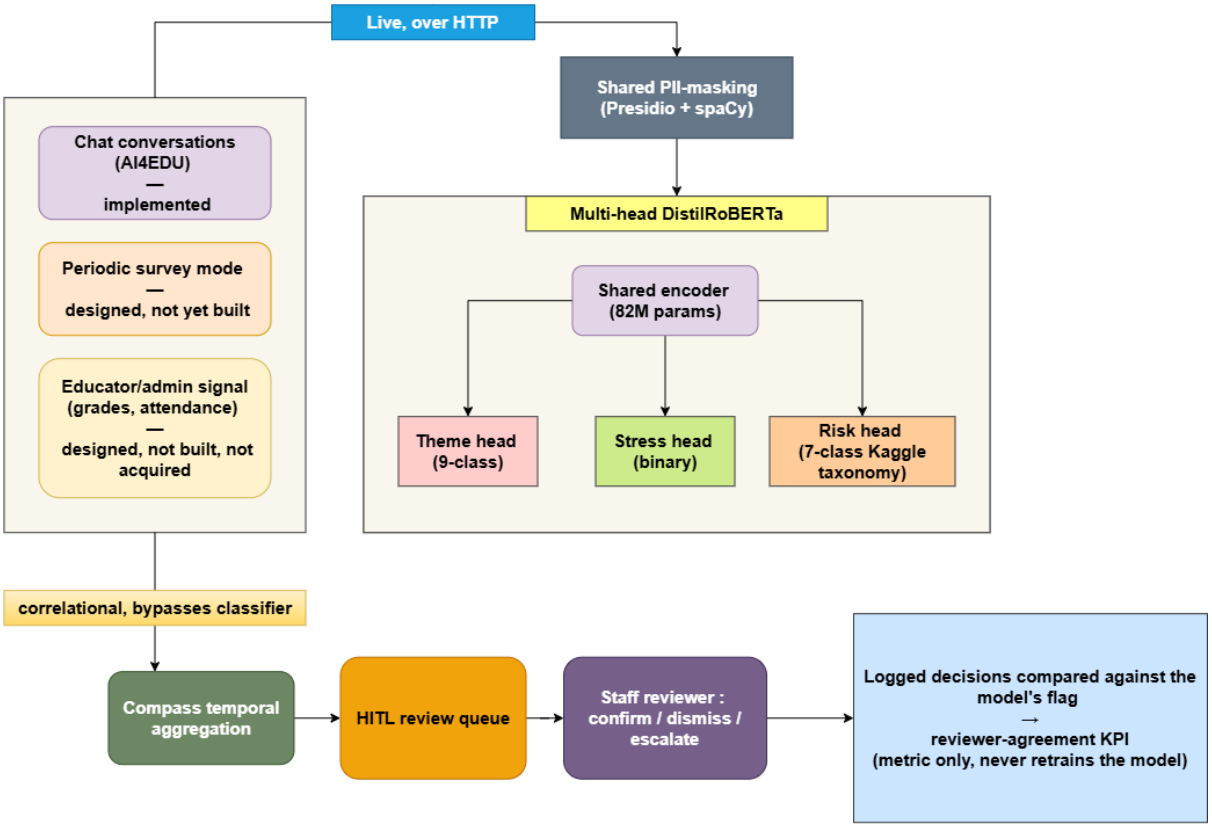


Figure 3.3: Deployed system architecture: training data, multi-head classifier, multi-modal input paths, PII masking, *Mental-Health Compass* aggregation, and the human-in-the-loop review queue. Dashed edges mark designed-but-not-implemented paths or evaluation-only feedback.

3.4 Privacy and Ethics Evaluation (RQ3)

RQ3 is evaluated in three parts. **RQ3.1 (privacy–utility ablation):** the performance cost of anonymization (PII removal before model input) and of context minimization (truncating the amount of student text available to the model), following the privacy-by-design methodology of Bao et al. [2023]. **RQ3.2 (on-premise vs. cloud-LLM comparison):** the on-premise fine-tuned classifier against prompted cloud LLMs (gpt-4o, gpt-4o-mini) on the same tasks and test manifests, evaluating macro-F1, latency, and data-exposure class together – the same on-premise, local-cloud rationale for distributed, resource-constrained systems argued by Haseeb [2025] for Industrial Internet of Things (IIoT) deployments, applied here to educational mental-health text. **RQ3.3 (ethics audit):** the resulting system is scored against the published ethics-assessment framework of Drira et al. [2024] — privacy (0–3), transparency reporting (0–10), replicability (0–5), interpretability (0–2), and fairness (binary) — the same instrument used to establish the field-level 65%/72%/18% gaps in Section 2.1.

Cloud-LLM prompt design. Every cloud cell uses the same fixed system/user prompt template across models and tasks, so that model and variant (zero-/few-shot, raw/masked) are the only things that change. The system message instructs the model to act as "a careful text classifier used in a research benchmark," to classify into exactly one of the given categories, and to respond with JSON only ({"label": "<category>"}, no explanation) — this constrains parsing without a schema-enforced Application Programming Interface (API) mode, and unparseable responses are counted explicitly (0–4 per cell, Section 4.3.2). The user message lists the task's categories with a short natural-language description (e.g., for risk: "the mental-health status expressed in the statement"), followed by the text to classify. The *few-shot* variant prepends one worked example per class, each drawn once from the training split (never the test split) at a fixed random seed, truncated to 400 characters, in the same Text: … / {"label": …} format as the query — so the model sees the exact output format demonstrated, not just described. All cloud calls use temperature 0 for reproducibility; no chain-of-thought or reasoning tokens are requested, since the task is a single-label classification, not open-ended generation. This is a deliberately simple, uniform, and untuned prompt design — the caveat in Section 4.3.2 that a stronger prompt or JSON-schema-constrained output might close some of the on-premise advantage on the risk task follows directly from this choice. The exact prompt templates are reproduced in Section A.5.

3.5 KPIs

- **Detection:** macro-F1 (primary), per-class precision/recall with high-risk-class recall reported separately, Receiver Operating Characteristic - Area Under the Curve (ROC-AUC), false-positive rate on real educational conversations (AI4EDU as "normal traffic").
- **Trend/Compass:** detection latency in days/weeks versus a single-message baseline, alert volume per student-week against a planned staff-capacity budget.
- **Responsible AI:** privacy-preservation effectiveness (performance delta under anonymization/minimization ablations) plus Drira et al. ethics scores; calibration via ECE and reliability diagrams; bias assessed only indirectly, via planned behavioral probes (dialect/length/style sensitivity, not yet run), since no demographic metadata exists in any usable corpus.
- **Efficiency:** latency per message, peak memory, parameter count, on commodity on-premise hardware.
- **System/HITL:** reviewer agreement, time-to-review, and usefulness ratings from planned counselor/psychologist sessions (not yet conducted); evidence is quoted directly rather than summarized, so RAG groundedness metrics do not apply (Section 3.3.4).

Chapter 4

Results

All seven planned models in the comparative benchmark (RQ2 , Section 3.3.1) are complete; the cloud-LLM comparison arm and the ethics audit (RQ3) are complete; RQ1's headline zero-shot evaluation against the 54-session human gold set is also complete (Section 4.1). All figures below are the mean of three seeds with documented, fixed-seed split manifests; every artifact (code, split manifests, per-run metrics) is in the project repository. Figure 4.1 maps how the three research questions relate: one shared v2 model feeds three independent evaluation branches.

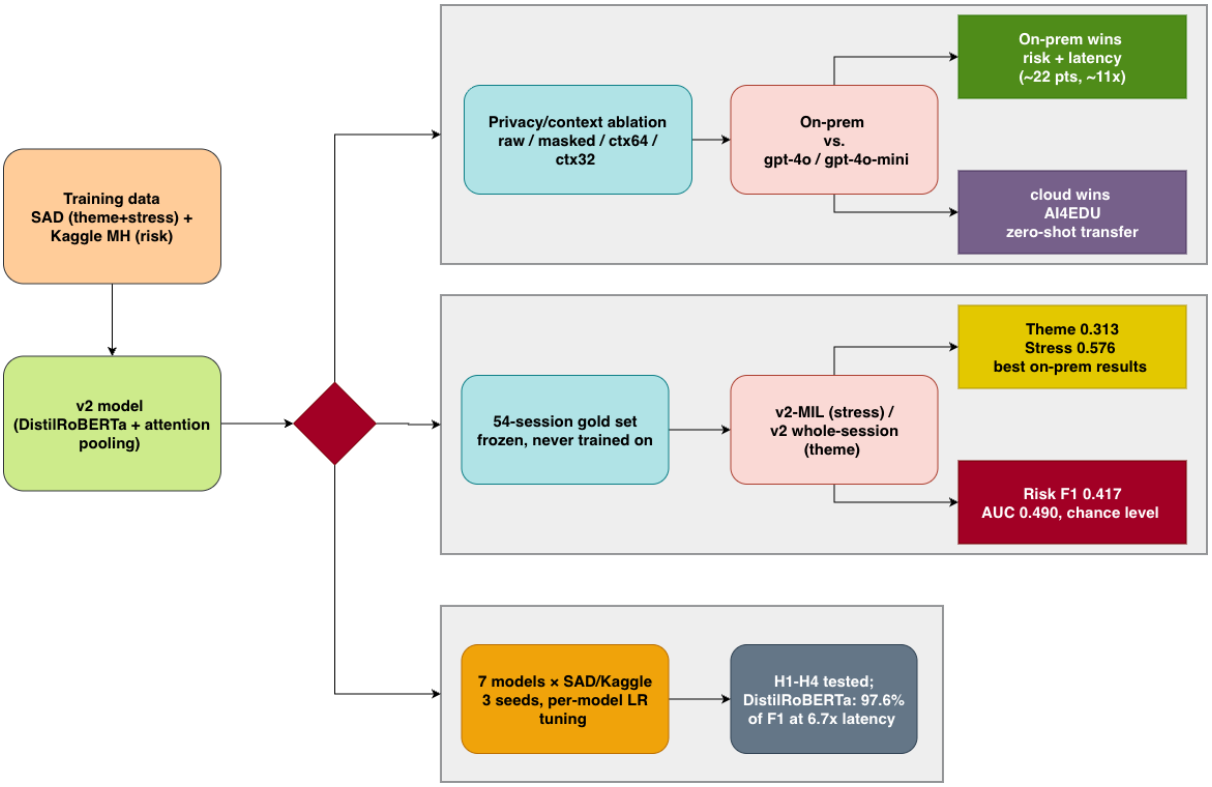


Figure 4.1: RQ1–RQ3 evaluation pipeline: a single v2 model, trained once on public proxy data, feeds three independent evaluation branches. Green nodes are positive/settled findings; red nodes are unresolved or negative findings, stated as plainly as the positive ones.

4.1 RQ1 — Zero-Shot Transfer to AI4EDU

The base v2 multi-task checkpoint was tested zero-shot against the held-out, never-trained-on 54-session AI4EDU gold set (Table 3.1). Table 4.1 summarizes the headline result across all three heads before the detailed variant-by-variant discussion below.

Table 4.1: RQ1 headline results: zero-shot transfer to the AI4EDU gold set.

Task	Result	Metric	Best variant
Theme	0.313	macro-F1	v2 whole-session
Stress	**0.576**	macro-F1	v2-MIL
Risk	0.490	ROC-AUC (chance level)	fails under every variant

A second variant, **v2-MIL**, changes only the *inference-time* input framing — it reuses v2's trained weights unchanged (the same base checkpoint, trained on SAD/Kaggle only) and encodes a session as a bag of turns rather than one whole-session pass, aggregating turn-level scores into one session-level score under the standard multiple-instance-learning framing (the specific pooling rule is fixed below; cf. Yang et al., 2016, Sun et al., 2019, Pappagari et al., 2019). This adds zero additional parameters and requires no additional training. Table 4.2 reports macro-F1 for v2-MIL on every dataset it was actually evaluated on; cells marked "not evaluated" are stated explicitly rather than left blank, since a variant's dataset coverage is itself part of the result (v2-MIL is a session-aggregation strategy, so single-statement corpora like SAD/Kaggle have nothing to aggregate and were never run through it).

Table 4.2: v2-MIL: fixed bag-of-turns aggregation over v2's frozen weights, no additional training.

Dataset (head)	v2-MIL
SAD / Kaggle (in-domain)	not evaluated — session-aggregation strategy; nothing to aggregate over single statements
AI4EDU gold (theme)	0.2296 — worse than v2 whole-session (0.3128); turn-level aggregation discards a session-level property
AI4EDU gold (stress)	**0.5758** — best on-prem AI4EDU result overall
AI4EDU gold (risk)	not tested — the 7-category-to-4-band mapping used for whole-session risk scoring was not wired into the turn-aggregation pipeline; even scored, only 3 of 54 sessions exceed "none"

Settled recommendation. v2-MIL is the recommended architecture for the stress head — best on-prem AI4EDU result of the whole investigation, at zero parameter cost over v2. Theme is best served by v2's whole-session encoding with no turn aggregation at all; v2-MIL makes theme worse, consistent with theme being a session-level property that per-turn aggregation

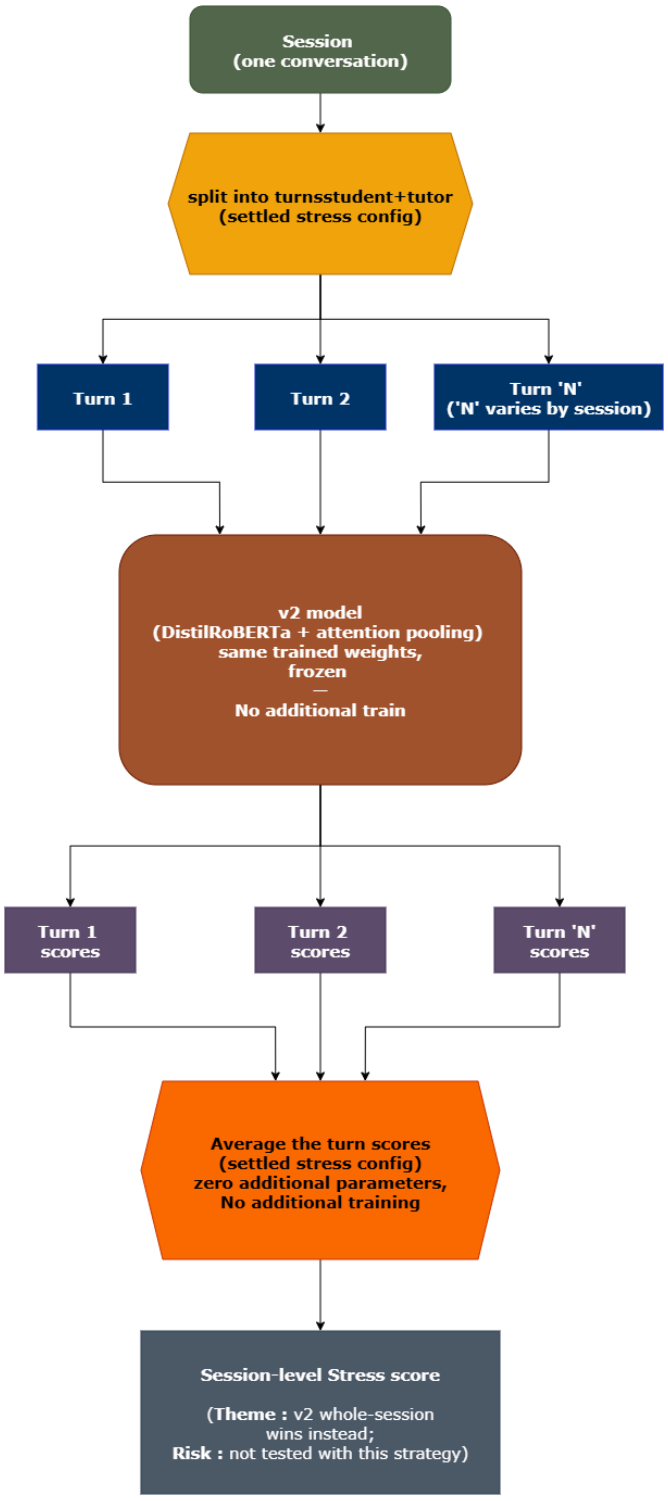


Figure 4.2: v2-MIL: a session is split into turns, each encoded by v2's unchanged weights, and turn-level scores are combined by fixed (untrained) aggregation — zero additional parameters.

discards. Risk remains unresolved on AI4EDU regardless of variant(AUC 0.490, chance level; full caveats in Section A.7.2).

v2-MIL's exact configuration is fixed to **student+tutor turns, probability-level aggregation, mean across turns** (0.5758). This is stated explicitly because it was selected as the best of eight candidate configurations (2 turn-scopes × 2 aggregation levels × 2 pooling functions) evaluated on the same 54-session gold set used to report the figure, with no separate development set on which to fix the aggregation hyperparameter beforehand — an additional, honest caveat alongside the single-positive-session caveat that already governs every stress number on this corpus (Section A.7.2). The theme recommendation is unaffected: v2 outperforms every one of the eight v2-MIL theme variants, so the choice does not turn on which variant is compared.

4.2 RQ2 — Comparative Benchmark

H1 confirmed — the published headline result does not survive rigorous evaluation. Khan et al. [2025] report RoBERTa-Large at 97% accuracy and, same-metric, a **93% macro-F1** on the Kaggle mental-health aggregate. That dataset contains 3.7% duplicate statements (1,721 exact + 220 near-duplicate); under our deduplicated, per-model-tuned, three-seed protocol, the same dataset yields **0.77–0.81 macro-F1** across all seven models (accuracy: 0.798–0.833) — a genuine 12–16 point collapse on the same metric the original paper itself reports, not an artifact of comparing accuracy to macro-F1. This directly supports the decision-support, non-diagnostic framing adopted in Section 1.1.

H2 confirmed — lightweight retention is now measured, not assumed. Table 4.3 reports macro-F1 retention against RoBERTa-Large (the largest, reference model) for all seven models, mean of three seeds. DistilRoBERTa (82M parameters) retains **97.6%/96.3%** of RoBERTa-Large's (355M) macro-F1 on risk/theme at **6.7× lower CPU latency** (12.1 vs 81.6 ms per message) — the first such measurement on mental-health text; prior work recycled GLUE-era efficiency figures. DistilBERT reaches 94.9% on risk, marginally below the pre-registered 95% threshold. High-stakes performance does not scale with model size: Suicidal-class recall is 0.78–0.83 across all seven models, with Mental-RoBERTa (not the largest model, RoBERTa-Large) best.

Table 4.3: Retention vs. RoBERTa-Large, single-task macro-F1 (mean of 3 seeds).

Model	Params	Risk	Theme	Stress*	CPU p50/msg
DistilBERT	66M	94.9%	96.5%	115.6%	12.3 ms
DistilRoBERTa	82M	**97.6%**	96.3%	116.9%	**12.1 ms**
BERT-base	109M	97.5%	97.6%	117.0%	26.5 ms
MentalBERT	109M	97.9%	96.7%	120.0%	29.8 ms
Mental-RoBERTa	125M	99.6%	99.4%	123.8%	26.4 ms
RoBERTa-base	125M	98.6%	97.9%	122.5%	26.7 ms
RoBERTa-Large	355M	100% (0.811)	100% (0.742)	100% (0.615)	81.6 ms

*Stress exceeds 100% because RoBERTa-Large's single-task stress fine-tune collapsed (per-seed 0.721/0.640/0.484 — a known large-model instability on small binary data); its multi-task run is stable at 0.776 ± 0.006 (Section 4.2.1).

Model selection. DistilRoBERTa is selected as the deployment candidate on four converging metrics from Table 4.3, not on accuracy alone: best distil-class retention (97.6% on the highest-stakes task), the lowest measured CPU latency of any model in the grid (12.1 ms), stable fine-tuning across all three tasks (unlike RoBERTa-Large), and the best expected-calibration-error profile of the distil class (Section 4.2.1). RoBERTa-base is the fallback if an additional 0.9–1.2 F1 points are judged to justify 2.2× the latency; RoBERTa-Large is judged not worth its 6.7× latency for at most 2.7 additional F1 points, and is also the least reliable of the seven to fine-tune. Table 4.4 confirms the retention gaps tested are statistically real where it matters (risk task, $p < 0.05$ for every pair tested against RoBERTa-Large), while the domain-pretraining question (H3, next) is not.

Table 4.4: McNemar significance, seed-1 test predictions (exact binomial).

Pair (task)	Accuracy A	Accuracy B	p
DistilRoBERTa vs RoBERTa-Large (risk)	0.8195	0.8331	1.2×10^{-4} — real but small
DistilBERT vs RoBERTa-Large (risk)	0.8063	0.8331	8.5×10^{-13}
RoBERTa-base vs RoBERTa-Large (risk)	0.8234	0.8331	4.0×10^{-3}
MentalBERT vs BERT-base (risk)	0.8195	0.8189	**0.90** — no difference (H3)
DistilRoBERTa vs RoBERTa-Large (theme)	0.7344	0.7627	2.2×10^{-2}
DistilRoBERTa single vs multi (risk)	0.8195	0.8141	0.071 — MTL cost n.s.

H3 confirmed — domain-specific pretraining buys little. MentalBERT is statistically indistinguishable from plain BERT-base (McNemar $p = 0.90$; deltas −0.7 to +1.9 macro-

F1), and Mental-RoBERTa adds only +0.8 to +1.1 over RoBERTa-base (not significant on risk, $p = 0.07$) — both well inside the ≤ 3-point margin their authors reported. The ~32 GPU-days of domain-adaptive pretraining behind these checkpoints is confirmed unnecessary for this task family, supporting fine-tuning released general-domain checkpoints over domain pretraining.

H4 confirmed — naive splits inflate scores. Holding sources out (group-aware splitting) drops theme by −3.1 points and stress by **−14.1** points — style leakage is real and worst on the highest-stakes signal family, which directly calibrates expectations for AI4EDU zero-shot transfer (RQ1).

4.2.1 Multi-Task Learning: Calibration and Stability

Multi-task learning — neutral on accuracy, valuable for calibration and stability. Table 4.5 reports the single- vs multi-task comparison across *all seven* benchmarked architectures — not only the distil class — that this finding rests on. Joint theme+stress+risk training changes macro-F1 by at most ~2.3 points in either direction (excluding RoBERTa-Large's stress stabilization, discussed below), but improves calibration (ECE) in **16 of 21 comparisons** — with the largest wins where single-task calibration was worst (theme: 0.114 → 0.083 DistilBERT, 0.112 → 0.090 DistilRoBERTa, 0.124 → 0.089 RoBERTa-base; stress: 0.051 → 0.028 DistilBERT, 0.043 → 0.029 DistilRoBERTa) — and, at full grid scale, unexpectedly **prevented RoBERTa-Large's fine-tuning collapse** on the binary stress task (single-task seeds 0.721/0.640/0.484, mean 0.615 ± 0.099; multi-task 0.776 ± 0.006), its single largest calibration and stability gain (ECE 0.134 → 0.029) of the entire grid. For a pipeline whose alerts aggregate confidence-weighted signals over time (Section 3.3.2), trustworthy confidence and training stability are more deployment-relevant than raw accuracy, so multi-task learning's value is reported as calibration and stability rather than an accuracy gain.

Table 4.5: Single- vs multi-task macro-F1 and ECE, all seven benchmarked architectures (test split, mean over 3 seeds).

Model	Task	Single F1	Multi F1	Single ECE	Multi ECE
DistilBERT	Theme (9-class)	0.7160 ± .0037	0.7182 ± .0030	0.114	**0.083**
	Stress (binary)	0.7111 ± .0112	0.7216 ± .0052	0.051	**0.028**
	Risk (7-class)	0.7697 ± .0016	0.7672 ± .0022	0.060	0.062
DistilRoBERTa	Theme	0.7143 ± .0058	0.7071 ± .0034	0.112	**0.090**
	Stress	0.7188 ± .0123	0.7169 ± .0133	0.043	**0.029**
	Risk	**0.7914** ± .0041	0.7830 ± .0020	0.055	**0.051**
BERT-base	Theme	0.7237 ± .0019	0.7274 ± .0052	0.072	0.073
	Stress	0.7197 ± .0091	0.7428 ± .0170	0.038	**0.030**
	Risk	0.7905 ± .0009	0.7770 ± .0010	0.074	**0.061**
MentalBERT	Theme	0.7172 ± .0032	0.7184 ± .0012	0.102	0.118
	Stress	0.7382 ± .0170	0.7193 ± .0101	0.040	**0.036**
	Risk	0.7936 ± .0013	0.7949 ± .0022	0.064	0.072
Mental-RoBERTa	Theme	0.7369 ± .0050	0.7324 ± .0021	0.099	**0.083**
	Stress	0.7613 ± .0067	0.7586 ± .0053	0.033	**0.029**
	Risk	0.8075 ± .0034	0.8033 ± .0055	0.059	**0.053**
RoBERTa-base	Theme	0.7259 ± .0026	0.7247 ± .0052	0.124	**0.089**
	Stress	0.7537 ± .0071	0.7723 ± .0064	0.034	**0.025**
	Risk	0.7999 ± .0036	0.8021 ± .0032	0.059	**0.051**
RoBERTa-Large	Theme	**0.7417** ± .0064	0.7266 ± .0010	**0.082**	0.097
	Stress[†]	0.6150 ± .0985	**0.7764** ± .0058	0.134	**0.029**
	Risk	**0.8109** ± .0042	0.8121 ± .0040	0.056	**0.050**

[†]Single-task RoBERTa-Large stress fine-tuning is unstable (per-seed 0.721/0.640/0.484); multi-task training incidentally stabilizes it, the largest calibration and stability gain in the table.

A follow-up Dreaddit training ablation, suggestive but not confirmed (n=1), is reported in Section A.6.

4.2.2 Trend-Data Feasibility

The timestamped Reddit Mental Health Dataset (Low et al. 2020, Open Data Commons Public Domain Dedication and License, ODC-PDDL; 577,033 posts across 13 subreddits selected for relevance to this thesis's own theme/risk taxonomy – e.g. *bipolarreddit* for the Bipolar risk class, *relationships/parenting/ divorce/personalfinance* for SAD's theme categories – January 2018 – April 2020) makes weekly post-volume trends per community directly computable, and contains 2,490 authors with ≥ 5 posts spanning ≥ 28 days — a sufficient cohort to demonstrate per-person rolling-window aggregation (the *Mental-Health Compass*) on real longitudinal data, complementing the synthetic survey-cadence timelines described in Section 3.3.2. Median posts per author is 1, so population-level trends remain the primary analysis; the longitudinal cohort is a demonstration set, not the main evaluation.

4.3 RQ3 — Privacy, Ethics, and Cloud Comparison

4.3.1 RQ3.1 — Privacy–Utility Ablation

No published mental-health study measures the F1 cost of anonymization or context minimization at all (Gap 5); Table 4.6 reports, to our knowledge, the first such measurement in this literature, re-measured on v2's architecture (attention pooling, 512 tokens) rather than the single-task setup this ablation originally used, so the finding is grounded in the architecture actually recommended elsewhere in this thesis. PII prevalence is 22% of SAD sentences and **59.5% of Kaggle posts** (including person names, phone numbers, and emails), making anonymization a materially load-bearing control rather than a formality.

Table 4.6: RQ3.1 privacy–utility ablation on v2: macro-F1 by arm (mean ± sd of 3 seeds).

Arm	Theme (SAD)	Stress (SAD)	Risk (Kaggle)
Raw baseline	0.7130 ± .0131	0.7137 ± .0081	0.8081 ± .0058
PII-masked	0.7047 ± .0039 (−0.8)	0.6956 ± .0116 (−1.8)	0.8045 ± .0037 (−0.4)
Context 64 tokens	0.7142 ± .0056 (+0.1)*	0.7061 ± .0186 (−0.8)*	0.7498 ± .0038 (**−5.8**)
Context 32 tokens	0.7160 ± .0077 (+0.3)*	0.7263 ± .0213 (+1.3)*	0.7090 ± .0047 (**−9.9**)

*SAD sentences are ≤30 words (median 12), so 64-token budgets barely truncate — these cells measure seed noise, not minimization (deltas fall within 1 sd); the informative context curve is the risk task (median 61 words, p90 277).

Anonymization costs 0 to −1.8 macro-F1 (theme −0.8, risk −0.4, stress −1.8), and context minimization is free on short chat/survey-style text (the SAD-style deltas are within seed noise) but costly on long posts (risk −5.8/−9.9 at 64/32-token budgets — a proxy-corpus artifact of the Kaggle data, not the target modality): short chat/survey turns, the deployment modality this thesis targets, have nothing to cut and lose nothing; long social-media posts lose 6–10 points under aggressive truncation. Stress detection leans on temporal/personal specifics ("my exam *tomorrow*") that DATE_TIME/PERSON masking removes, while theme classification survives because topical vocabulary ("exam," "family") is not PII — a plausible mechanism for where the small stress-task cost comes from. The pattern is the same shape as the original single-task measurement (anonymization cheap, short-text context-minimization free, long-text context-minimization costly), confirming the masking cost is a property of the data, not the pooling architecture. **RQ3.1's core answer: privacy-by-design does not meaningfully degrade detection.**

4.3.2 RQ3.2 — On-Premise vs Cloud-LLM Comparison

All 14 cells are complete (theme/stress/risk × {zero-shot, few-shot} × {raw, masked} for gpt-4o-mini, plus gpt-4o zero-shot raw/masked for risk; $n = 7{,}611$ for risk, $n = 1{,}024$ for theme/stress). Table 4.7 compares the deployed multi-task v2 model against the best cloud-LLM cell per task, both evaluated on the same PII-masked test text (3-seed mean for v2, Table 4.6).

Table 4.7: Deployed multi-task v2 (on-premise) vs best cloud-LLM variant, both on PII-masked test text, macro-F1.

Task	On-prem v2 (masked)	Cloud best (masked)	Detail
Theme (9-class)	0.705	**0.716** (4o-mini few-shot)	near-tie, cloud ahead ~1 pt
Stress (binary)	0.696	**0.742** (4o-mini zero-shot)	cloud ahead ~5 pts
Risk (7-class)	**0.805**	0.582 (4o-mini few-shot)	**on-prem wins by ~22 pts**

Notably, the smaller and cheaper gpt-4o-mini beats gpt-4o on *every* matched risk-task variant — few-shot examples help more than model scale here — and masking costs almost nothing on the cloud side (raw vs. masked cells differ by 1–2 points throughout). Suicidal-class recall is slightly higher on-premise (0.828) than for the same masked cloud cell (0.813). Latency is 578–699 ms (API, p50) versus 54.1 ms for the quantized v2 model on CPU, all three heads jointly (~11–13× faster); unmasked cloud use would additionally send text of which 59.5% contains PII to a third party. Computed from actual prompt token counts, the entire cloud arm cost

approximately $7.65 to run; per 1,000 risk-task messages, gpt-4o-mini costs ~$0.022 and gpt-4o ~$0.366, against $0 marginal cost on-premise. **RQ3 conclusion (in-domain test data, both arms masked): the deployed on-premise model is close to cloud on theme, trails it on stress, and is vastly more accurate on the task the pipeline actually escalates on (risk), at roughly an order of magnitude lower latency and zero marginal cost.**

Two caveats are stated explicitly: cloud cells are single-pass temperature 0, no seed variance) with one reasonable but untuned prompt design, so a stronger prompt or JSON-schema setup might close some of the risk-task gap; and ECE is not comparable across arms since the API does not expose calibrated probabilities.

4.3.3 Deployment Footprint: Quantization, Latency, and Cost

The deployment candidate (multi-task DistilRoBERTa, Section 4.2) is exported to ONNX and INT8-quantized for commodity, GPU-free on-premise hardware. Table 4.8 reports the quantization cost per task head, measured against the same fp32 PyTorch checkpoint on held-out test data (release manifest and validation report committed with the artifact).

Table 4.8: ONNX INT8 quantization: accuracy cost and argmax agreement vs. fp32.

Head	n	fp32 macro-F1	INT8 macro-F1	Argmax agreement
Theme	1,024	0.7087	0.7016	96.4%
Stress	1,024	0.6857	0.6814	99.4%
Risk	7,611	0.8015	0.8023	97.7%

Quantization shrinks the on-disk model from 326 MB (fp32) to **82 MB (INT8)** — about a 4.0× reduction — at a macro-F1 cost of at most −0.71 points (theme; risk is effectively free, +0.08), with 96–99% argmax agreement against the fp32 reference across all three heads. Measured on CPU (no GPU), the quantized model runs **all three heads jointly at p50 54.1 ms / p95 64.8 ms per message** — higher than DistilRoBERTa's single-head figure in Table 4.3 (12.1 ms) because this checkpoint uses a 512-token window (vs. 128 for the Stage A benchmark models), not because of the extra task heads; still comfortably within the near-real-time budget defined in Section 1.4 (minutes, not milliseconds, are required), which leaves orders of magnitude of headroom.

Table 4.9 places the full deployment picture — accuracy, latency, and marginal cost — side by side for the on-premise INT8 model and the two cloud arms evaluated in Section 4.3.2.

The on-premise, quantized path dominates on every axis for the task the pipeline actually

Table 4.9: Deployment comparison: on-premise INT8 vs. cloud LLM (risk task, the highest-stakes comparison). Both arms use raw (unmasked) text; Table 4.7 reports the corresponding masked-vs-masked comparison.

	On-prem DistilRoBERTa (INT8)	gpt-4o-mini	gpt-4o
Risk macro-F1	**0.802–0.810**	0.592 (best variant)	0.574
Latency (p50/msg)	**54.1 ms** (all 3 heads)	578–699 ms	578–699 ms
Hardware	commodity CPU, no GPU	third-party API	third-party API
Marginal cost/message	**$0**	~$0.022/1,000 msgs	~$0.366/1,000 msgs
Data leaves institution	**no**	yes (raw: 59.5% contains PII)	yes (raw: 59.5% contains PII)

escalates on: it is markedly more accurate, roughly 10–13× faster, free per message at inference time (amortized compute only, no per-call fee), and never sends student text off-site. This is the concrete evidence behind the on-premise argument in Section 3.4 and the load-bearing premise of the whole privacy-by-design pipeline (Section 3.3.5).

4.3.4 RQ3.3 — Ethics-Framework Audit

Table 4.10 scores the system against the Drira et al. instrument [Drira et al., 2024], alongside the field-level baseline that motivated Gap 2 and Gap 4 (Section 2.1).

Table 4.10: Ethics-framework audit (Drira et al. 2024 instrument).

Dimension	Our score	Field baseline
Privacy (0–3)	**3/3**	65% of 59 studies score 0
Transparency reporting (0–10)	~8/10 (unverified exact items)	hyperparameter reporting 30%, validation-metric reporting 49%
Replicability (0–5)	**5/5**	rare, given the 65%/72% privacy/ethics gaps
Interpretability (0–2)	1/2 (deliberate scope decision)	only 2/59 use any XAI
Fairness (binary)	partial, not a clean pass	18% (11/59) report demographics

- **Privacy (3/3):** Anonymization is met, and its performance cost is measured (Section 3.4) — going beyond what the framework asks (applying a technique) into what it says the field never does (measuring the cost). Consent is met, inherited from AI4EDU's own collection-time process (CC-BY-4.0). AI4EDU is used for zero-shot evaluation only (RQ1) and never enters model training, and the labeling effort itself — human annotators and an LLM weak-labeler both reading real student–tutor conversations to build the gold set — is covered by the same public license.

- **Replicability (5/5), the strongest-evidenced dimension:** A full reproducibility re-

run, covering the entire benchmark across two passes (2026-07-20, 2026-08-30), independently confirmed 51 of 53 experiment/task/mode cells within ±0.02 macro-F1 tolerance — including H1's headline result — with the two exceptions traced to a previously documented model-instability issue and a modest overshoot (Section 3.3.1), on top of public code, fixed-seed committed split manifests, and scripted, documented data fetch for every dataset — a meaningfully stronger bar than the framework's minimum (public code alone), worth stating as exceeding the criterion rather than merely meeting it.

- **Interpretability (1/2), deliberately partial:** Explainability/XAI was deprioritized per an explicit scope decision (Section 1.2.1); what the system does provide is interpretable output by construction — theme labels, calibrated confidence (ECE-measured), and traceable source-message references — not post-hoc explanation.

- **Fairness — the one honest shortfall:** The field's own bar is low (are demographics reported at all — met by 18%); this thesis reports a coarse country/cohort breakdown for AI4EDU (Ireland/Greece/Cyprus) but no systematic age, gender, or grade reporting, and no demographic-subgroup bias evaluation has actually been run. The flat Suicidal-class recall across model sizes (0.78–0.83) is a model-size robustness check, not a demographic-fairness check, and is not cited as evidence for the latter. **Honest verdict: partial credit, not a clean pass** — the concrete gap to close is a systematic extraction of AI4EDU's self-disclosed age/grade distribution.

Net read. The system is genuinely strong on privacy-cost measurement and replicability — both essentially absent field-wide — while fine-grained fairness reporting remains the one honest open item, stated here rather than glossed over.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

The reviewed literature establishes that transformers and LLMs can detect emotional and thematic signals in text reliably enough to support — though never replace — human judgment. It equally establishes what the field has not done: no study analyzes educational conversational data, none aggregates signals over time, privacy and ethics are recommended far more often than implemented (65% of studies include no privacy element), human oversight is endorsed but not designed, and efficiency claims are recycled rather than measured (Chapter 2).

This thesis addresses these gaps with a privacy-first well-being decision support, architected around temporal aggregation and human review as core design principles. The key findings are:

- **RQ2 (H1).** A rigorous protocol collapses a published headline result from 0.93 macro-F1 to 0.81 macro-F1 on the same data — with every other benchmarked model landing in the same range 0.77–0.81 range
- **RQ2 (H2).** A lightweight distilled model retains 96–98% of a 4× larger model's performance at a fraction of the latency.
- **RQ2 (H3).** Domain-specific pretraining buys at most 1–2 points.
- **RQ2 (H4).** Naive train/test splitting inflates scores by up to 14 points on the highest-stakes task (Section 4.2) — all of which calibrated expectations for the AI4EDU zero-shot transfer evaluation (RQ1, Section 4.1), now complete against the 54-session human gold set.

- **RQ1 — architecture.** The originally deployed architecture (v1) failed on real conversation for two diagnosable, architectural reasons — truncation at 128 tokens and single-position CLS pooling — and fixing both (v2: attention pooling, 512 tokens) improves every in-domain head and narrows, without closing, the gap to gpt-4o on theme and stress.

- **RQ1 — v2-MIL.** A further, zero-cost refinement on top of v2 closes more of that gap for stress specifically: **v2-MIL** (a session-as-bag-of-turns inference strategy, zero additional parameters, no additional training) is the best on-premise stress result found; theme is best served by v2's whole-session encoding, where turn-level aggregation only makes it worse.

- **RQ1 — risk transfer.** Risk transfer to AI4EDU remains unresolved (AUC 0.490, chance level) regardless of architecture — a negative finding reported plainly rather than a threshold left ambiguous.

- **RQ3.** The privacy and cloud-comparison arm (Section 4.3) shows that anonymization is nearly free, that on-premise inference beats every evaluated cloud LLM variant on the highest-stakes in-domain classification task (risk) by roughly 20 points at 1/10th the latency and zero marginal cost — even though gpt-4o wins on AI4EDU zero-shot theme and stress, the reverse comparison — and that the system scores strongly on privacy-cost measurement and replicability against a published ethics framework, while candidly leaving fine-grained (rather than country-level) fairness reporting as the one open item.

The intended outcome is not a diagnostic instrument but a deployable, lawfully operable (General Data Protection Regulation, GDPR- and research-ethics-compliant) early-signal tool that strengthens the existing student-support workflows of European schools — where the cost of a missed struggling student is high, and the cost of an untrustworthy alert is a system nobody uses. Its temporal-aggregation (*Mental-Health Compass*) and HITL review mechanisms are demonstrated as designed architecture, with end-to-end validation on real student data left as the natural next step.

5.2 Limitations

1. **Domain shift is the core limitation, now precisely characterized.** Training data is social-media text (Reddit/Twitter aggregates); the real student-conversation evaluation corpus (Table 3.1) is genuine educational conversation but curricular in nature, so well-being signals are expected to be rare. Detection is therefore evaluated on proxies and false-positive behavior on real conversations; LLM-generated synthetic distress data was considered as a stress-test corpus but scoped out of this thesis (Section 3.1.1).

2. **English-only scope; Greek-language sessions explicitly excluded.** The Greek-language Greece/Cyprus student-conversation sessions (Table 3.1) are not processed: the PII-masking step (Presidio) and the classifier stack are both English-only, and this is a deliberate scoping decision rather than a tooling gap to be closed. This follows the field's dominant convention (MentalBERT itself is English-only and defers multilingual coverage to future work); machine-translation was considered and rejected for this thesis, since translationese is itself a style-leakage risk — consistent with this thesis's own finding that splitting by source rather than at random measurably degrades detection (Section 4.2), evidencing that models key on surface style as well as content.

3. **No Swedish conversational data is public**; Swedish-language validity is not pursued in this thesis. The NLV survey sample obtained so far is too small to support a classifier role, a data-volume limitation pending a larger sample rather than a decision against the modality (Section 3.1.1).

4. **Label quality.** Public-dataset mental-health labels are weak (subreddit-derived) and LLM-generated weak labels inherit LLM biases; mitigated by psychologist validation of the taxonomy and a human-coded gold set used for the real-conversation zero-shot evaluation (Section 4.1).

5. **Scale.** The real-conversation corpus in use (Table 3.1) supports evaluation and thematic analysis, not training — comparable in scale to prior proof-of-concept qualitative studies in this space.

6. **No demographics** exist in any usable dataset, so fairness evaluation is limited to behavioral probes and dataset-bias analysis rather than full demographic-parity claims (Section 4.3.4) — stated openly rather than papered over.

7. **Ethical/legal constraints.** Any new Swedish student-data collection requires *etikprövning* regardless of consent, and minors require parental consent; this risk is managed by making all confirmatory analyses runnable on already-public data.

8. **Text-only.** Linguistic signals are an incomplete window on well-being; every flag is an input to human judgment, never a conclusion.

9. **No end-user evaluation.** Neither the escalation rules nor the HTIL review layer has been evaluated with the people who would use them: no counselor, educator, or student has used the review queue, and no escalation rule has been scored against labelled outcomes. The applied contribution is therefore a working design whose usefulness in real student-support workflows remains to be measured (Section 5.3).

5.3 Future Work

- Acquiring genuine, session-level, real student-conversation training data remains the open constraint on further closing the theme/stress gap to gpt-4o, and on resolving risk transfer, which is unresolved at chance level regardless of architecture.

- Extend the cloud-LLM comparison with a tuned prompt or JSON-schema-constrained output to test whether the ~20-point on-premise advantage on the risk task narrows under a stronger prompting setup.

- Counselor/psychologist walkthrough sessions of the review-queue prototype, to turn the qualitative-usefulness KPI (Section 3.5) from planned into measured.

Appendix A

Supplementary Methodology Detail

A.1 Additional Data Sources Explored

The itemized list below is the full search trail behind Section 3.1.2's methodology paragraph.

- **Conversational.** No public, downloadable, verifiably high-school-age corpus exists beyond AI4EDU. Crisis Text Line and the Trevor Project are gated/internal-only; *ConvCounsel* and RECIPE4U/StudyChat are real and downloadable but university-level, with no mental-health taxonomy fit.
- **Survey.** YRBSS and PISA are high-school-age and downloadable but closed-form only (no free text) — useful for base-rate calibration, not NLP. GLSEN's open-ended quotes exist only as aggregate PDF reports. A single-school COVID-anxiety survey (PMC9524706, $n = 155$) was inspected directly: its "why" fields are multi-select checkboxes, not narrative text. HBSC (WHO Health Behaviour in School-aged Children, 5 rounds 2001–2018, ~50 countries) was requested and received (~310MB, individual-level microdata) but is closed-form only, like YRBSS/PISA, and was decided out of scope entirely – not used for classifier training/eval or for the *Mental-Health Compass*.
- **Reddit teen-identified.** No candidate (r/teenagers research, RedDust, a JMIR classifier, a 2026 Drexel preprint) is age-verified, and none release downloadable post+comment-thread data. Scraping r/teenagers directly via PRAW is the only remaining path (self-identified/unverified age); **not started**, pending API credentials and a go/no-go decision.
- **eRisk-parallel per-user timestamped datasets.** SMHD/RSDD/RSDD-Time are **dead** (Georgetown pulled the distribution form, citing Reddit API ToS changes); UMD

Reddit Suicidality is **parked**. eRisk access, initially blocked, was subsequently obtained and used as a supplementary stress-head fine-tuning source (Track C, Section 3.1.1). Our own Low et al. Reddit corpus already implements the per-author, per-timestamp cohort idea (≥5 posts/≥28-day span), just without a per-user risk label.

- **RedditESS and Code RED.** Both reviewed in full; neither releases a downloadable conversational dataset (post-IDs/code only) — citable as related work only.

None of the items above are wired into any pipeline; each carries its concrete status (dead, parked, obtained, not started) rather than an undifferentiated "future work."

A.2 Deduplication Parameter Justification

Table A.1: Deduplication design choices and their rationale.

Choice	Reason
MinHash-LSH, not all-pairs	All-pairs Jaccard on the larger corpus is ~1.4 billion comparisons; LSH is $\sim O(n)$
Character 3-grams	Robust to typos and small edits in short, informal text; word-level shingles break on a single changed word
64 permutations, 16 bands of 4 rows	Banding threshold $\approx(1/16)^{1/4}\approx 0.5$, so pairs at the true Jaccard 0.9 threshold are caught with probability >0.9999
Verify with true Jaccard	LSH surfaces candidates, not decisions; exact verification removes false positives
Threshold 0.9	Conservative: removes near-copies (reposts, trivial edits) without merging posts that merely share phrasing

A.3 Selected Hyperparameters

Table A.2 reports the learning rate selected by the per-model sweep (Section 3.3.1) for every (model, task) cell in the comparative benchmark, including the multi-task variant. Val-split sweep scores per candidate LR are in each run's sweep_summary.json, not reproduced here.

A.4 Model Artifact Detail

The attention-pooling mechanism (Section 3.3.5) adds a negligible parameter cost over the base encoder: 82,132,242 → 82,133,011 (+769 parameters). The released classifier is release/multihead-distilro (ONNX int8, ~78 MB; fp32, ~326 MB).

Table A.2: Selected learning rate per model and task (validation-split sweep, seed 42).

Model	Theme	Stress	Risk	Multi (all 3 heads)
DistilBERT	3×10^{-5}	2×10^{-5}	3×10^{-5}	3×10^{-5}
DistilRoBERTa	5×10^{-5}	2×10^{-5}	5×10^{-5}	3×10^{-5}
BERT-base	1×10^{-5}	3×10^{-5}	5×10^{-5}	2×10^{-5}
MentalBERT	5×10^{-5}	3×10^{-5}	3×10^{-5}	5×10^{-5}
Mental-RoBERTa	3×10^{-5}	3×10^{-5}	3×10^{-5}	2×10^{-5}
RoBERTa-base	3×10^{-5}	1×10^{-5}	3×10^{-5}	2×10^{-5}
RoBERTa-Large	2×10^{-5}	3×10^{-5}	2×10^{-5}	1×10^{-5}

A.5 Cloud-LLM Prompt Templates

Section 3.4's Cloud-LLM prompt design paragraph describes this template at a summary level; the exact text is reproduced here for reproducibility. Every cloud cell (any model × task × variant) uses the identical system message:

> You are a careful text classifier used in a research benchmark. Classify the text into exactly one of the given categories. Respond with JSON only: {"label": "<category>"} — no explanation.

The user message lists the task's categories with a short natural-language description, then the text to classify. For the risk task:

> Categories (the mental-health status expressed in the statement): ["Anxiety", "Bipolar", "Depression", "Normal", "Personality disorder", "Stress", "Suicidal"]
>
> Text: {text}

The *few-shot* variant inserts one worked example per class between the category list and the query text, sourced only from the training split (never test), one random draw per class at fixed seed 7, each example truncated to 400 characters:

> Categories (...): [...]
>
> Text: {example text, truncated to 400 chars}
> {"label": "{example label}"}
>
> Text: {example text, truncated to 400 chars}

{"label": "{example label}"}

... (one example per class)

Text: {query text}

All calls use temperature 0 and a 30-token output budget (sufficient for the required JSON label, not for chain-of-thought). Source: src/wellbeing/llm_eval.py (SYSTEM, build_prompt, fewshot_from_train) and src/wellbeing/ai4edu_llm_eval.py (same system message; no few-shot support, since the AI4EDU gold set has no training split to draw examples from).

A.6 Dreaddit Cross-Task Transfer Ablation

A follow-up ablation, separate from the main single- vs multi-task comparison (Section 4.2.1), tested whether adding Dreaddit (an external stress-annotated Reddit corpus) to stress training affects other tasks. Adding Dreaddit's training split to stress training left in-domain SAD-test performance flat (within noise) while lifting Dreaddit-test score substantially (+12.3 to +20.7 single-task, +8.6 multi-task DistilRoBERTa — now ~96% of MentalBERT's own published anchor). In the multi-task setting, risk performance also improved by +1.1/+2.0 F1, even though risk's own training data (Kaggle) never changed — a shared-encoder cross-task transfer effect distinct from the calibration finding in Section 4.2.1. This is an n=1 ablation and is flagged as suggestive, not confirmed.

A.7 Reddit Trend-Corpus Face-Validity Check

A supplementary check tested whether the eRisk-fine-tuned checkpoint (Track C) produces plausible severity-discrimination on real long-form text, using the timestamped Reddit Mental Health Dataset [Low et al., 2020] (13 subreddits chosen for relevance to this thesis's taxonomy, 577,033 posts, 2,490 authors) as a face-validity testbed – the corpus carries no labels of any kind, so subreddit identity (crisis communities assumed more severe than everyday ones) is the only available, unverified proxy for severity. The originally deployed classifier fails this check: stress rate is near-uniform (~97%) regardless of subreddit, a 7.6-point spread. The eRisk-fine-tuned checkpoint passes: crisis subreddits (ptsd, healthanxiety, depression, anxiety,

suicidewatch, mentalhealth) score 83–86% versus 23–39% for mundane ones (personalfinance, relationships, parenting), a 63.2-point spread that stays above 50 points across a decision-threshold sweep from 0.5 to 0.95 – and specifically fixes *bipolarreddit*, the originally deployed classifier's single worst-ranked subreddit. Under this thesis's own v2-MIL mean-aggregation rule the effect is attenuated but not washed out (34.9-point spread, same correct ranking); under a stricter any-turn-positive rule it largely disappears. This is a face-validity check, not an accuracy measurement, and its predictions are not used as a *Mental-Health Compass* severity signal pending a dedicated calibration study.

sectionAI4EDU Weak-Labeler Selection

Before either candidate model touched any real student text, the weak-labeler was selected via a local, on-premise bake-off run on public proxy validation data only, never AI4EDU (n=1,000/task: SAD's validation split for theme and stress, Kaggle's validation split for risk, since SAD carries no risk labels) – consistent with this thesis's privacy-by-design principle (Section 4.3) of minimizing which systems are exposed to student-adjacent content. Qwen2.5-32B-Instruct was selected over Llama-3.3-70B despite being 2× smaller.

Table A.3: AI4EDU weak-labeler bake-off: F1 by task.

Model	Theme	Stress	Risk
Qwen2.5-32B-Instruct	0.728	0.741	0.226
Llama-3.3-70B	0.674	0.732	0.228

A.7.1 AI4EDU Gold-Set Composition

The 54-session gold set combines 19 stratified sessions (guaranteeing coverage of every theme class, both stress-positive sessions, and the single risk-flagged session) with 35 additional randomly sampled sessions. Risk labels are especially sparse within this set: only 3 sessions are labeled "low," with no "medium" or "high" sessions present at all.

A.7.2 AI4EDU Evaluation Caveats

Risk's sparse, narrow label range (Section A.7.1) is too few positives and too limited a class spread for a stable macro-F1, which is why risk transfer is reported via ROC-AUC instead (0.490, chance level). The recommended v2-MIL configuration for stress (student+tutor turns, probability-level aggregation, mean across turns) was itself selected as the best of eight candi-

dates evaluated on this same 54-session set, with no separate development set available – an additional, honest caveat alongside the single-positive-session fragility that governs every stress and risk number on this corpus.

A.8 Code and Data Availability

The experimental codebase (data pipelines, training and evaluation scripts, and the RQ1 zero-shot evaluation harness) is maintained at https://github.com/mh-compass/mh-models, and the *Mental-Health Compass* site implementation at https://github.com/mh-compass/compass-web. Both repositories are private; access can be requested from the authors at ennc0d3@gmail.com.

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